

Downstream Effects of U.S. Biofuel Policies

Jungkeon Jo, Xiaoli Etienne, and Michael K. Adjemian*

This version: 9/19/2026

Abstract

We study how U.S. biofuel policy shocks transmit to downstream markets, including food, livestock, energy, and overall consumer prices. Relying on the institutional structure of U.S. biofuel policy and high-frequency price movements around regulatory announcements, we construct a novel biofuel policy news series and use it as an external instrument in a structural VAR framework. A stringent biofuel policy shock leads to significant inflation across both upstream and downstream markets. In response to the shock, biofuel, oil, and field crop prices rise; in the downstream market, consumer food prices and the headline CPI increase, while household food expenditures fall. Fats and oils and dairy prices rise more sharply than other food categories. In livestock markets, cattle and beef prices rise as feed costs increase, and feedlot placements contract in the long run. The shock also raises retail energy prices and consumer inflation expectations. Food and electricity prices respond more for lower-income than for higher-income households. While higher commodity prices, driven by biofuel policy, can support agricultural producer revenues, consumers finance these gains through higher retail prices for both fuel and food, and lower-income households bear a disproportionate burden.

Keywords: Biofuel policy, Renewable Fuel Standard, high-frequency identification, external instrument SVAR, food price inflation, price transmission, consumer prices, renewable diesel

JEL classification: Q42, Q18, E31, C32, Q11, Q48

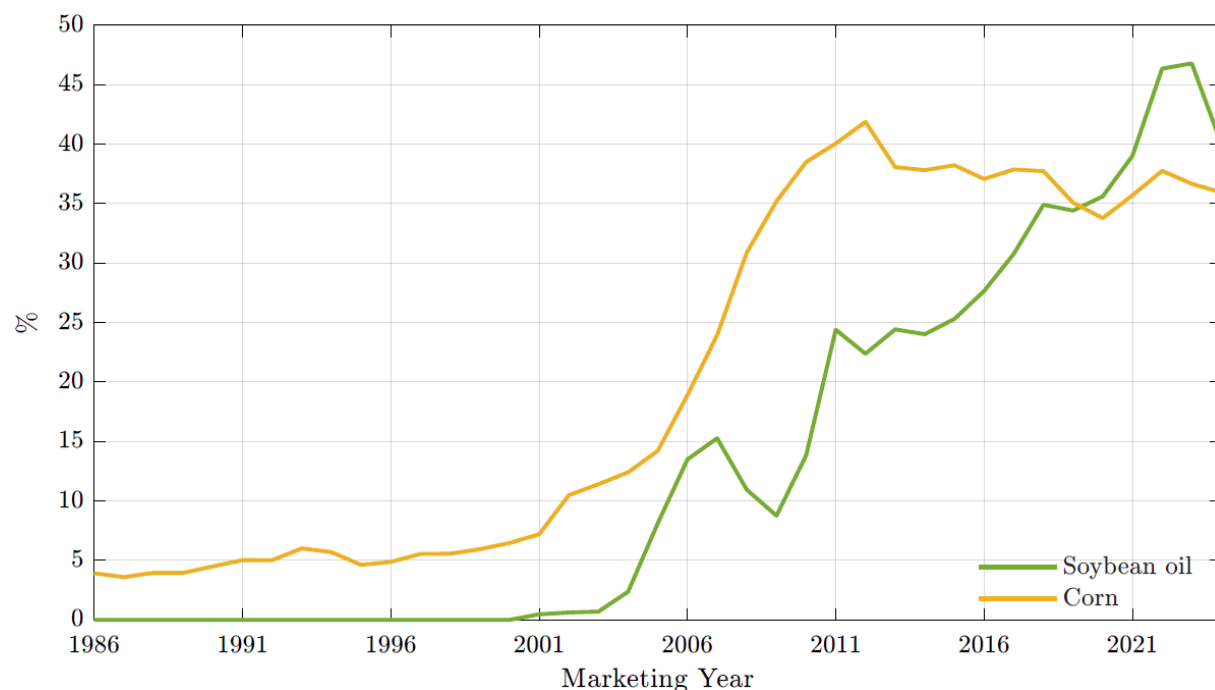
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1. Introduction

Energy and food prices are among the most politically sensitive components of inflation, directly affecting consumers through gasoline and grocery expenditures. One federal policy in the United States linking the two is the Renewable Fuel Standard (RFS), which mandates minimum volumes of renewable fuels blended into the U.S. transportation fuel supply (EPA, 2025a). By expanding demand for agricultural commodities, the mid-2000s biofuel boom supported farm incomes and rural development through higher commodity prices and investment (Rajagopal and Zilberman, 2007; Moschini and Lapan, 2012).

Today, biofuels represent a primary outlet for U.S. agricultural commodities, with the scale of this dependence growing rapidly over time. As illustrated in Figure 1, over the last decade, 35-40 percent of U.S. corn production has been allocated to ethanol manufacturing. At the same time, the biodiesel and renewable diesel sectors have absorbed a rapidly growing share of oilseed output, with biofuel use of soybean oil surging from less than 1 percent in 2001/02 to over 40 percent by 2024/25 (Avileis, 2025).

Figure 1. Shares of U.S. corn for ethanol and soybean oil for biofuel production



Source: Table 5. 'Corn supply, disappearance (million bushels), and share of total corn used for fuel alcohol (percent)' & Table 6. 'Soybean oil supply, disappearance (million pounds), and share used for biofuel (percent)' in USDA Bioenergy Statistics.

This structural transformation has important implications for downstream food and energy markets. Under the RFS, obligated parties, primarily refiners and importers, must acquire and retire sufficient Renewable Identification Numbers (RINs) annually to comply with mandated blending volumes. As Lade and Bushnell (2019) show, RFS mandates impose implicit taxes on fossil fuels and implicit subsidies for biofuels, with both adjusting endogenously in response to policy stringency. Policy actions that tighten enforcement or increase compliance costs can affect fuel prices and alter demand for agricultural feedstocks, creating potential spillovers to food markets (de Gorter and Just, 2009; Lade and Bushnell, 2019).

These linkages motivate a central question. To what extent do policy-driven shifts in biofuel demand propagate and feed into broader inflationary pressures? This question is directly relevant to the longstanding “food versus fuel” debate, or whether diverting agricultural commodities toward energy production raises food prices and threatens food security (von Braun et al, 2008; Hochman et al., 2011; Zilberman et al., 2013). Proponents of biofuel mandates argue that price effects are modest and temporary, as supply responses and productivity gains offset demand pressures (de Gorter and Just, 2010). Critics counter that even modest increases in commodity prices translate into meaningful food price inflation for vulnerable consumers, with disproportionate burdens on low-income households who spend larger shares of their budgets on food and energy (Wright, 2011; 2014).

Previous research has documented significant effects of biofuel policies on upstream agricultural commodity markets. During the initial expansion of corn ethanol production following the 2005 RFS, biofuel demand was identified as a key driver of corn price movements (Carter et al., 2011; Carter et al., 2017). Studies of the expanded 2007 RFS mandate further show that biofuel policies affected both the level (Hochman, Rajagopal, and Zilberman, 2010; Hausman, Auffhammer, and Berck, 2012; Chen and Khanna, 2013) and volatility of corn prices (Wright, 2011; 2014). A parallel strand of research examined the relationship between agricultural commodity prices and energy prices, documenting significant price linkages and volatility spillovers between the two markets (Zhang et al., 2009, 2010; Serra, 2011; Serra, Zilberman, and Gil, 2011, among others). More recent studies have begun to examine the renewable diesel boom, focusing on the

soybean oil and crushing market (Avileis, 2025; Avileis, Swanson, and Irwin, 2026; Kang and Lee, 2025).

This literature, however, shares three important limitations. First, most empirical contributions focus on upstream commodity markets rather than downstream food and energy prices faced by consumers. This distinction is important because commodity price movements do not translate one-for-one into retail prices—processing margins, transportation costs, and retailer pricing behavior can absorb or amplify shocks as they move through the supply chain (Meyer and von Cramon-Taubadel, 2004; Nakamura and Zerom, 2010). Moreover, retail prices are the outcomes most directly relevant to household welfare, especially for low-income households. Although recent work has examined pass-through of RFS implicit subsidies to retail fuel prices (Lade and Bushnell, 2019) and impacts on beef herd size (Davis, Adjemian, and Langemeier, 2023), less is known about how it affects broader measures of domestic retail food and energy prices.

Second, establishing the causal effects of biofuel policy is challenging because observed price movements reflect many simultaneous forces rather than isolated policy changes. Most existing studies rely on reduced-form models or partial equilibrium models. Reduced-form approaches that use biofuel production volumes or mandate levels as policy proxies face endogeneity concerns because policy decisions may respond to prevailing market and economic conditions (de Gorter and Just, 2010; Lade and Bushnell, 2019; Irwin, 2019). Partial equilibrium models, while useful for counterfactual analysis, require assumptions about market clearing, supply elasticities, and cross-market linkages that may be difficult to validate, particularly during periods of rapid market changes. These limitations motivate an identification strategy that separates exogenous policy variation from other contemporaneous market shocks.

Third, aside from a few recent studies (Avileis, 2025; Avileis, Swanson, and Irwin, 2026; Kang and Lee, 2025), relatively little empirical evidence exists on the effects of recent renewable diesel boom. As Figure 1 shows, this expansion has shifted the center of gravity in biofuel feedstock use from corn toward soybean oil. As a result, the price effects of biofuel policy in this newer setting remain largely uncharacterized.

This paper addresses these gaps by developing a high-frequency measure of biofuel policy news. We construct the news series from changes in ethanol and soybean oil futures prices around regulatory announcements by the Environmental Protection Agency (EPA) and the California Air Resources Board (CARB), as well as *Reuters* reports of leaked EPA regulatory decisions. We then use the series as an external instrument in a Proxy Structural Vector Autoregression (SVAR-IV) framework (Stock and Watson, 2012; Mertens and Ravn, 2013) to estimate the dynamic effect of biofuel policy shocks on upstream and downstream markets.

This design draws on the logic of high-frequency identification, which uses asset price movements around well-timed announcements to isolate market-relevant policy news. The approach was pioneered in the monetary policy literature using interest rate surprises around Federal Open Market Committee (FOMC) meetings (Bagliano and Favero, 1999; Kuttner, 2001; Faust et al., 2004; Gürkaynak et al., 2005; Gertler and Karadi, 2015; Nakamura and Steinsson, 2018; Bauer and Swanson, 2023, among others) and has since been extended to oil markets (Känzig, 2021), U.S. Treasury markets (Phillot, 2025), and agricultural commodity markets (Adjemian and Jo, 2025). We bring this identification strategy to the biofuel regulatory setting. Policy announcements occur on discrete, observable dates, and active futures markets.

Our baseline SVAR model includes the global real Brent oil spot price alongside a suite of U.S.-specific variables: the real biofuel price, the real field crop price, household food spending, the headline consumer price index (CPI), and the food-at-home CPI. The latter three variables represent the key downstream price indices, capturing aggregate food expenditures, overall inflation, and grocery prices, respectively. The biofuel price, a production-weighted composite of ethanol and biodiesel prices, is used to measure biofuel market shocks. The sample spans November 2007 to May 2026 at monthly frequency. Because it includes the extreme macroeconomic volatility of early 2020, we follow Lenza and Primiceri (2022) and down-weight COVID-19 observations, preventing a handful of pandemic months from dominating the residual covariance structure and distorting the estimated dynamics. Section 4 describe the adjustment in detail.

We find that tighter biofuel policy generates inflationary pressures that pass from upstream commodity markets to downstream consumer prices. The biofuel policy shock raises biofuel, oil, and field crop prices significantly. These cost pressures propagate gradually to consumers, raising consumer prices and food prices. Household food spending declines by a magnitude comparable to the food price increase, suggesting that consumption adjusts roughly one-for-one.

The inflationary burden is not evenly distributed across food categories. Commodities most directly tied to biofuel production, particularly fats and oils and dairy, experience the sharpest price increases, while categories with limited feedstock exposure show muted responses. Producer prices respond more strongly than consumer prices across several categories, indicating that upstream cost pressures are partially absorbed before reaching consumers. Beyond food markets, the shock raises retail fuel and oil prices and consumer inflation expectations, indicating that biofuel policy affects not only realized prices but also beliefs about future inflation.

The burden also falls unevenly across households. Using a distributional consumer price index (Jaravel, 2026) constructed separately by income group, we find that a biofuel policy shock raises food and electricity prices more for lower-income than for higher-income households. Since lower-income households devote a larger share of their budgets to food and energy, the resulting increase in their cost of living is larger.

We also assess the quantitative importance of these shocks. Historical decompositions (HDs) indicate that biofuel policy has contributed to consumer price movements, and accounting for a nontrivial portion the recent inflation episode. Forecast error variance decompositions (FEVDs) confirm these findings, explaining a large share of biofuel price variation and a long-run variation in field crop prices, consumer food prices, and the headline CPI.

This paper contributes to the literature in three ways. First, we extend the focus of the food-versus-fuel literature beyond upstream agricultural commodity markets by estimating the effects of biofuel policy shocks on retail food prices and consumer food expenditures, which are central to evaluating the household-level consequences of biofuel policy. Second, we provide the first exogenous measure of U.S. biofuel policy shocks, which addresses the endogeneity problem

and can be used by other researchers. Third, we provide the first evidence on the distributional incidence of biofuel policy. Our results show that price increase in food and electricity is larger for lower-income households than for higher-income households.

The rest of this paper proceeds as follows. Section 2 provides institutional background on U.S. biofuel policy. Section 3 describes the identification strategy and the construction of the biofuel policy news series. Section 4 presents the empirical framework. Section 5 reports the main results. Section 6 presents robustness checks, and the last section concludes.

2. Institutional Background

2.1 The Renewable Fuel Standard

The Renewable Fuel Standard (RFS) mandates minimum annual volumes of renewable fuels in the U.S. transportation fuel supply. Established by the Energy Policy Act of 2005 and expanded under the Energy Independence and Security Act of 2007, the program was designed to reduce lifecycle greenhouse gas emissions, strengthen domestic energy independence, and support rural economies by creating stable demand for agricultural feedstocks. The statute envisioned renewable fuel use rising from 9 to 36 billion gallons between 2008 and 2022 (EPA, 2025b), though the EPA-determined renewable fuel obligation for 2023 reached 20.8 billion gallons (EPA, 2023a).

The EPA administers the RFS through a market-based system of Renewable Identification Numbers (RINs), tradeable compliance credits generated when biofuels are blended with petroleum-based fuels. Obligated parties, primarily refiners and importers, must acquire and retire sufficient RINs annually across four fuel categories, with the credit market allowing parties in deficit to purchase from those generating surplus (Gerveni and Irwin, 2025; Lade and Smith, 2025). Each year, the EPA sets Renewable Volume Obligations (RVOs), which translate statutory targets into percentage standards determining how many RINs obligated parties must retire. Proposed and final RVO announcements therefore convey regulatory news that directly shifts expected demand for RINs and biofuels.

Figure 2 plots RIN prices for biomass-based diesel (D4) and conventional renewable fuel (D6, primarily corn ethanol) from 2010 to 2025. The series reflects successive shifts in regulatory stringency.

Biodiesel mandate expansion (2010—2013). Biomass-based diesel mandates grew from 0.5 to over 1.3 billion gallons between 2009 and 2013 (EPA, 2025c), pushing D4 RIN prices up as production capacity struggled to keep pace. Prices eased as capacity expanded.

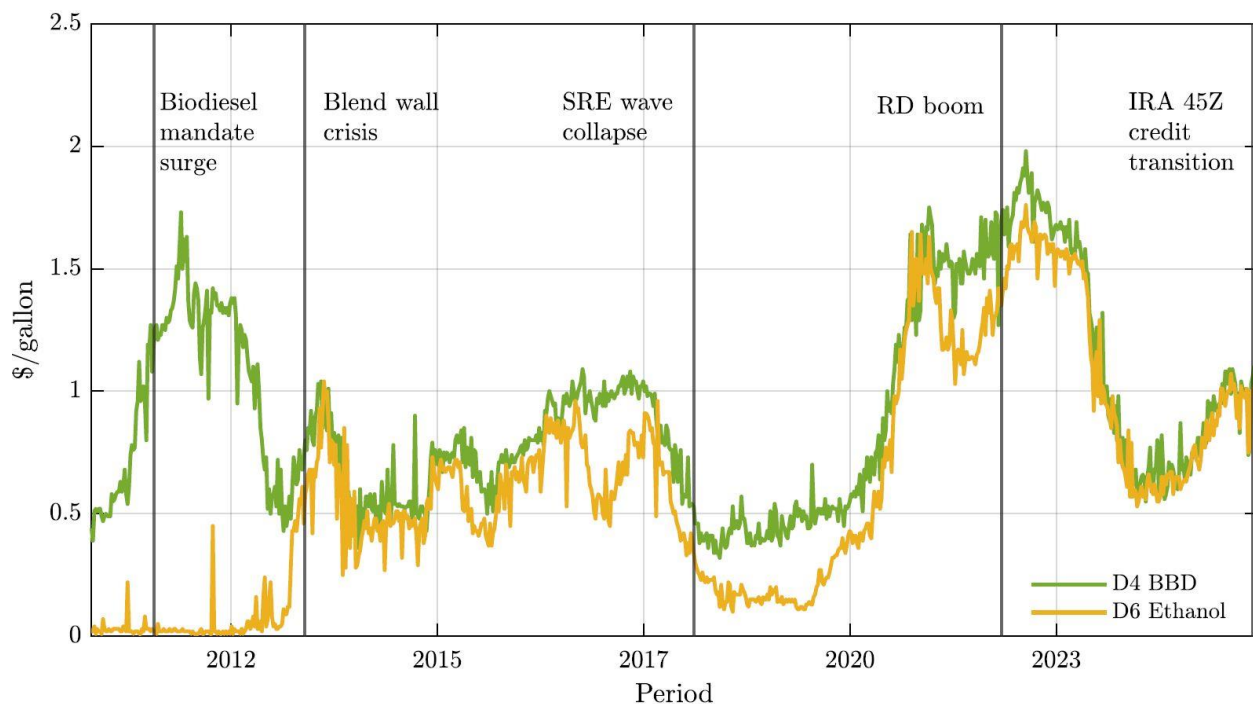
The binding blend wall (2013—2015). D6 RIN prices surged above \$1.00 per gallon as corn ethanol production approached the practical blending limit of roughly 10 percent (Lade and Smith, 2025). Unable to expand ethanol consumption further, obligated parties turned to higher-cost biodiesel, pulling D4 RIN prices up as well (Irwin and Good, 2013).

SRE wave collapse (2016—2019). An expanded use of small refinery exemptions reduced aggregate compliance obligations, weakening RIN demand and depressing both D4 and D6 prices (Irwin, 2019).

Renewable Diesel boom (2020—2023). D4 RIN prices reached historic highs as renewable diesel, a drop-in fuel without blend wall constraints, rapidly displaced conventional biodiesel (Ramsey, 2024; Gerveni, Hubbs, and Irwin, 2024).

IRA 45Z credit transition (2025—present). The Clean Fuel Production Credit under the Inflation Reduction Act introduced regulatory uncertainty, and both D4 and D6 prices declined as producers awaited clarity on qualifying fuel pathways (Buffie, 2025).

Figure 2. Biomass-Based Diesel (D4) and Renewable Fuel (D6) RIN prices, 2010-2025



Source: U.S. Environmental Protection Agency, RIN Trades and Price Information

Notes: Figure shows weekly volume-weighted average RIN prices for Biomass-Based Diesel (D4) and Renewable Fuel (D6) by transfer date, averaged across vintages when multiple production years were traded.

2.2 Low Carbon Fuel Standards (LCFS)

Complementing federal mandates, state-level Low Carbon Fuel Standards (LCFS) have become significant drivers of demand for lower-carbon advanced fuels. California implemented an LCFS in 2011 targeting a 10 percent reduction in transportation fuel carbon intensity by 2020, a target the California Air Resources Board (CARB) has since strengthened to 30 percent by 2030 (CARB, 2024). Unlike the RFS's volumetric mandates, the LCFS assigns credits and deficits based on lifecycle carbon intensity (CARB, 2025). Because California accounts for roughly 10 percent of U.S. transportation fuel consumption, and because Oregon, Washington, and New Mexico have adopted similar programs, the LCFS generates spillover effects nationally (Bushnell et al., 2024). CARB publishes regular LCFS compliance reports on credit generation, deficits, fuel volumes, and feedstock use, which convey market-relevant information about the balance of low-carbon fuel supply and compliance demand.

3. Identification Strategy

The key challenge in identifying biofuel policy effects is that changes in biofuel prices may reflect multiple simultaneous factors rather than policy alone. We address this by measuring biofuel futures price changes within a narrow window around regulatory announcements, when policy news is plausibly the dominant source of new information. The institutional setting, combined with active futures markets for biofuel energy, allow us to construct a high-frequency measure of biofuel policy news, which we use as an external instrument in the SVAR-IV framework described in Section 4.

3.1 Biofuel policy announcements

We construct a database of biofuel policy announcements from three primary sources, following Avileis, Swanson, and Irwin (2026). EPA announcements consist of proposed and final RVOs for the RFS.¹ Although these do not occur on fixed calendar dates, they are closely monitored because they determine expected compliance demand for RINs and biofuels. CARB announcements are quarterly LCFS compliance reports covering credit generation, deficits, fuel volumes, and feedstock consumption, which reveal the balance between low-carbon fuel supply and compliance demand. We supplement these institutional releases with *Reuters* articles reporting leaked or anticipated EPA regulatory actions, as *Reuters* has been a primary outlet for such information (Avileis, Swanson, and Irwin, 2026).

Because CARB reports are released at 3 pm PST, after market close, the relevant trading day is the following business day, as it is for *Reuters* articles published outside trading hours. When multiple biofuel announcements occur within a two-day window, we retain only the earliest announcement to avoid double-counting. In addition, none of our trading days coincide with the release dates of OPEC meetings or USDA crop inventory reports. The final sample comprises 113 announcements from November 2007 to May 2026, listed in Appendix A.

¹ EPA biofuel news are obtained from the ‘News, Notices and Announcements for the Renewable Fuel Standard’ EPA website: <https://www.epa.gov/renewable-fuel-standard/news-notice-and-announcements-renewable-fuel-standard>. There are different types of announcements. We focus on the publication date of ‘Proposed Renewable Fuel Standards’ and ‘Final Renewable Fuel Standards’ related to the supply of biofuel energy (see Avileis, Swanson, and Irwin, 2026, for details).

3.2 Biofuel futures markets

Ethanol and biomass-based diesel are the two primary biofuel commodities relevant to RFS compliance. For ethanol, we consider the Chicago Ethanol (Platts) Futures contract traded on NYMEX, the most liquid ethanol derivative in the United States (Gerveni et al., 2023). For biomass-based diesel, the CME biodiesel contract lacks sufficient price history and liquidity over our sample, so we use CBOT soybean oil futures as a proxy. Soybean oil is the dominant feedstock for U.S. biodiesel and renewable diesel production, accounting for about 55 percent of feedstock inputs (Gerveni, Hubbs, and Irwin, 2024), and soybean oil and biodiesel prices are closely linked through this relationship (Irwin and Good, 2017).

3.3 Identification of biofuel policy news

Following Känzig (2021), we measure biofuel policy news as the log change in biofuel futures prices within a daily window around each announcement:

$$IndividualBiofuelPolicyNews_{m,d,i}^h = 100 \times (\ln F_{m,d,i}^h - \ln F_{m,d-1,i}^h), \quad (1)$$

where d , m , and i are the day, announcement month, and commodity (ethanol or soybean oil), and $F_{m,d}^h$ is the settlement price for h -deferred futures contract. We consider $h=1,2$, the nearby and the first-deferred futures contracts.²

For each commodity, we aggregate across the two maturities using the first principal component (Pearson, 1901; Jolliffe and Cadima, 2016), which accounts for 78 percent of the variation. We then construct a single biofuel policy news measure by weighting each commodity's series based on monthly production shares:

$$BiofuelPolicyNews_{m,d}^h = \sum_i IndividualBiofuelPolicyNews_{m,d,i}^h \times S_{m,i} \quad (2)$$

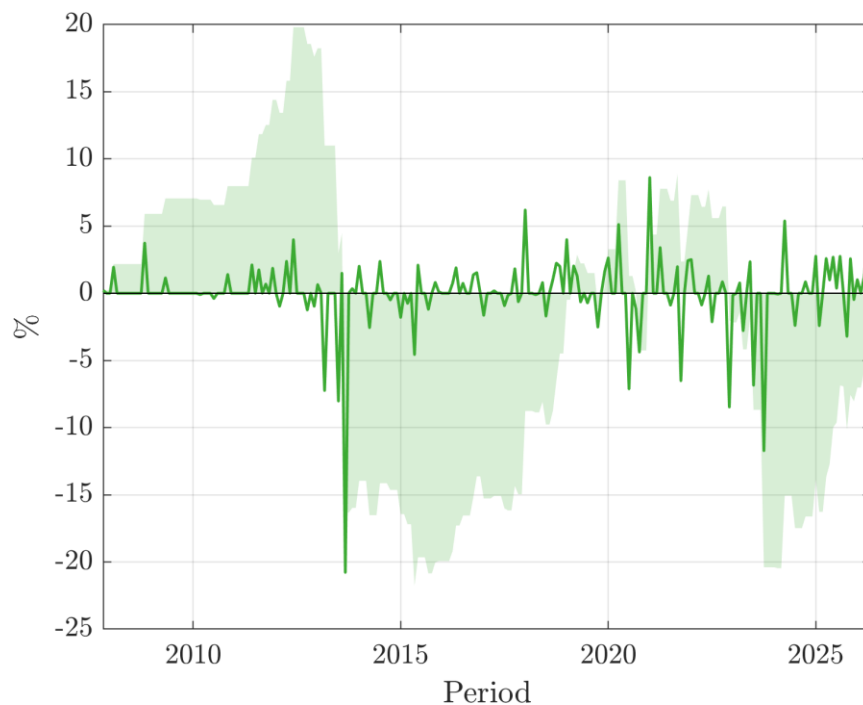
where $S_{m,i}$ is the production share of commodity i in month m .

Because the baseline is monthly, we aggregate the daily series accordingly. The monthly value equals the announcement-day news in months with a single announcement, the sum of announcement-day news in months with multiple announcements, and zero otherwise.

² A constant risk premium is assumed within the window.

Figure 3 illustrates the historical series of the U.S. biofuel policy news from November 2007 to May 2026. Biofuel policy news routinely surprises markets, especially in 2013-2015 and after 2020, periods that coincide with major expansions in biofuel production capacity. Surprises were more muted between 2016 and 2019, when small refinery exemptions reduced net compliance obligations and weakened the policy's aggregate demand impact (Coppess and Irwin, 2018; Irwin, 2019).

Figure 3. Biofuel policy news series, November 2007- May 2026



Source: Bloomberg terminal and authors' calculations

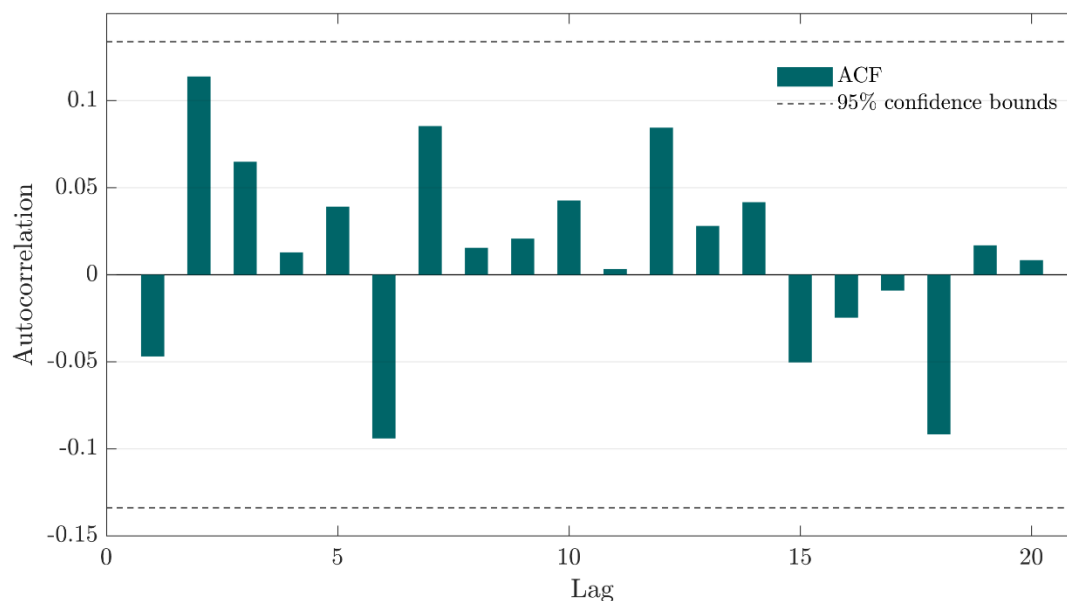
Notes: The biofuel policy news series is constructed as the first principal component of changes in the U.S. biofuel energy futures (ethanol and soybean oil; relative to their share of domestic production). The solid line represents the monthly biofuel policy news series, and the shaded area illustrates the cumulative series, which is the sum of the monthly news series over time.

3.4 Diagnostic tests of biofuel policy news

We verify the validity of the news series as an instrument through three diagnostic tests. Figure 4 shows that the instrument is not predictable from its own past. Table 1 reports Granger causality test results, showing that neither lagged baseline model variables nor biofuel production predict the news series.

We additionally evaluate informational sufficiency following Mori and Peersman (2026). A valid proxy should not be predictable from information available to market participants beyond that already embedded in the VAR. We examine whether four information blocks—financial and credit conditions, policy uncertainty, commodity and global factors, and monetary policy—predict the news series, the reduced-form residual of the real biofuel price equation, and the identified structural shock, after orthogonalizing each with respect to the baseline variables. The monetary policy shock predicts the raw news series, but the predictability disappears once the outcome is conditioned on the baseline model, and no block rejects after conditioning in any panel. The isolated per-series rejections that remain fall within what chance would produce across the full battery. As a robustness check, we re-estimate the model with an instrument orthogonalized against the federal funds rate. Appendix C reports the results.

Figure 4. Autocorrelation test



Source: Bloomberg terminal and authors' calculations

Table 1. Granger causality test

Variables	P-value
Biofuel policy news	0.7461
Real biofuel price	0.3094
Real oil price	0.9599
Real field crop price	0.4343

CPI	0.2391
Food-at-home price	0.9177
Household food spending	0.1670
Food service price	0.9688
Ethanol production	0.2655
Biodiesel production	0.8911
All	0.3075

Notes: The table reports p-values from Granger causality tests of whether each variable Granger-causes the biofuel policy news series. The lag order is 12, consistent with the baseline VAR specification. “All” reports the p-value for the joint test that all variables have not predictive power for the news series.

4. Empirical Approach and Data

We use the biofuel policy news series as an instrument in an SVAR-IV framework, following Stock and Watson (2012) and Mertens and Ravn (2013). This approach addresses the endogeneity of observed biofuel price movements, which may reflect policy actions as well as broader macroeconomic, agricultural, and energy-market conditions. The instrument isolates the policy-driven component by requiring the news series to be correlated with the structural shock of interest (“relevance”) but orthogonal to other disturbances (“exogeneity”). Unlike a direct shock measure, the instrument need not perfectly capture the structural shock since the news series captures only part of regulatory activity, is subject to measurement error (Stock and Watson, 2018), and may have limited statistical power (Nakamura and Steinsson, 2018).

4.1 SVAR model

We specify a reduced-form VAR (p) model:

$$y_t = \alpha + A_1 y_{t-1} + \dots + A_p y_{t-p} + u_t, \quad (3)$$

where y_t is an $n \times 1$ vector of endogenous variables comprising the real biofuel price, real oil price, real field crop price, CPI, food-at-home CPI, and household food spending; u_t is a vector of reduced-form innovations with covariance $Var(u_t) = \Sigma$; α is a vector of constants; and $A_1, \dots, A_p$ are coefficient matrices.

The reduced-form innovations in (3) are assumed to be linear combinations of structural shocks:

$$u_t = B\varepsilon_t, \quad (4)$$

where B is a non-singular structural impact matrix and ε_t is a vector of mutually uncorrelated structural shocks (i.e., $\text{Var}(\varepsilon_t) = \Omega$ is diagonal). From this mapping,

$$\Sigma = B\Omega B'. \quad (5)$$

We order the biofuel policy news shock first, reflecting the assumption that biofuel policy announcements respond to regulatory and political processes rather than to contemporaneous movements in energy, commodity, or consumer prices.³ Our focus is on identifying the structural impact vector b_1 or the first column of B , which traces how biofuel market news shocks affect the other variables in the system.

4.2 External instrument approach

Full identification of B requires n^2 restrictions, while Equation (5) provides only $n(n+1)/2$ unique restrictions. Rather than imposing additional zero or sign restrictions, we use an external instrument to partially identify b_1 alone. Let z_t denote the biofuel policy news series.

Identification of b_1 , requires

$$\mathbb{E}[z_t \varepsilon_{1,t}] = \theta \neq 0, \quad (6)$$

$$\mathbb{E}[z_t \varepsilon_{2:n,t}] = 0, \quad (7)$$

where $\varepsilon_{1,t}$ is the biofuel policy news shock and $\varepsilon_{2:n,t}$ collects the remaining structural shocks.

Under these two conditions, b_1 is identified up to a scaling factor (Stock and Watson, 2018):

$$\tilde{b}_1 \equiv \theta_0 \propto \mathbb{E}[z_t u_{2:n,t}] / \mathbb{E}[z_t u_{1,t}], \quad (8)$$

³ Note that this ordering refers to the structural shocks, not the endogenous variables in the VAR system. The first endogenous variable in y_t is the real biofuel price, a production-weighted composite of ethanol nearby futures and biodiesel spot prices, which serves as the observable measure of biofuel market conditions. The structural shock is normalized so that it raises the real biofuel price by 4 percent.

where θ_0 indicates unit-effect normalization and $\mathbb{E}[z_t u_{1,t}] \neq 0$.⁴

With unit-effect normalization, we scale the structural shock $\varepsilon_{1,t}$ such that it corresponds to a 4 percent increase in the real biofuel price, allowing the impulse responses to be interpreted directly in policy-relevant units (Stock and Watson, 2018):

$$\tilde{b}_1 \equiv \mathbb{E}[z_t u_{2:n,t}] / \mathbb{E}[z_t u_{1,t}] \times \omega. \quad (9)$$

4.3 COVID-era volatility weighting

Our sample spans the COVID-19 period, when unusually large macroeconomic disturbances could dominate the reduced-form covariance matrix and distort the estimated impulse responses. Following Lenza and Primiceri (2022), we model the reduced-form residuals as $u_t = s_t \cdot \varepsilon_t$, where ε_t is homoskedastic and the scale s_t equals one outside the pandemic window. Estimating s_t by maximum likelihood yields a scale of 18.72 on March 2020 and unity thereafter. The adjustment therefore amounts to heavily down-weighting March 2020, so that the VAR, estimated by weighted least squares with weights $1/s_t$, is not distorted by pandemic volatility.

4.4 Variables and Data

The baseline model includes six variables spanning the transmission pathway from biofuel policy through energy and commodity markets to consumer prices. The biofuel price, a production-weighted composite of the ethanol nearby futures price and the wholesale biodiesel price, is the policy-sensitive variable used to normalize the structural shock. Crude oil prices capture the transportation fuel channel, where biofuels and petroleum products are both complements and substitutes. Field crop prices, a production-weighted average of corn and soybean prices, capture upstream feedstock competition between food and fuel uses. Household food spending, the headline CPI, and food-at-home prices measure downstream pass-through to consumers.

⁴ Unit-effect normalization fixes the scale of $\varepsilon_{1,t}$ such that a unit increase in the structural shock leads to a one-unit increase in $y_{1,t}$. For instance, a 1 percent biofuel policy shock raises the biofuel price by 1 percent. This normalization has an important advantage over the more common unit standard deviation normalization: it allows direct estimation of dynamic causal effects in policy relevant units, making the impulse responses immediately interpretable without additional scaling (see Stock and Watson, 2018, for details).

The sample spans from November 2007 to May 2026. The start date corresponds to the first EPA announcement regarding the RFS. Appendix Table B provides full variable definitions, including those used in the extended and distributional analyses, and Table 2 reports summary statistics.

Table 2. Summary statistics, November 2007-May 2026

Variable	Obs.	Mean	Std. Dev.	Min	Max
Biofuel price (cents/gallon)	223	207.66	50.50	116.02	339.74
Oil price (cents/barrel)	223	78.30	23.99	26.85	133.59
Field crop price (cents/bushel)	223	633.17	150.12	445.73	1011.94
Consumer Price Index (CPI)	223	254.89	34.89	210.83	333.98
Food-at-home price index	223	251.05	32.48	205.12	320.84
Household Food Spending (Bil. Chained 2017 \$)	223	1009.27	131.39	835.70	1285.40

Sources: Bureau of Economic Analysis (BEA), Bureau of Labor Statistics (BLS), Federal Reserve Economic Data (FRED), Bloomberg terminal, USDA-ERS, USDA-NASS, and USDA Bioenergy Statistics

5. Results

In this section, we present empirical findings. The first-stage robust F-statistic is 12.60, above the conventional threshold, indicating that weak instrument concerns are limited, and instrument exogeneity is supported by the diagnostic tests reported in Section 3.4.

5.1 Impulse response functions

Figure 5 displays impulse responses to a biofuel policy shock, normalized to raise the real biofuel price by 4 percent, corresponding to a tightening of policy stringency.⁵ The vertical axis represents percentage changes, as all variables are expressed in natural logarithms. We also plot the 68 and 90 percent confidence intervals from a residual-based moving-block bootstrap with 10,000 bootstrap replications (Jentsch and Lunsford, 2022). The lag order is 12, consistent with the monthly data frequency.

⁵ We normalize the shock to a 4 percent increase in the real biofuel price, which approximates the impact response to a one-standard-deviation biofuel policy shock in the unnormalized model (approximately 4.4 percent). This choice keeps the magnitude of the shock economically interpretable while remaining close to a typical shock. The impulse responses from the unnormalized model are provided in Appendix D.

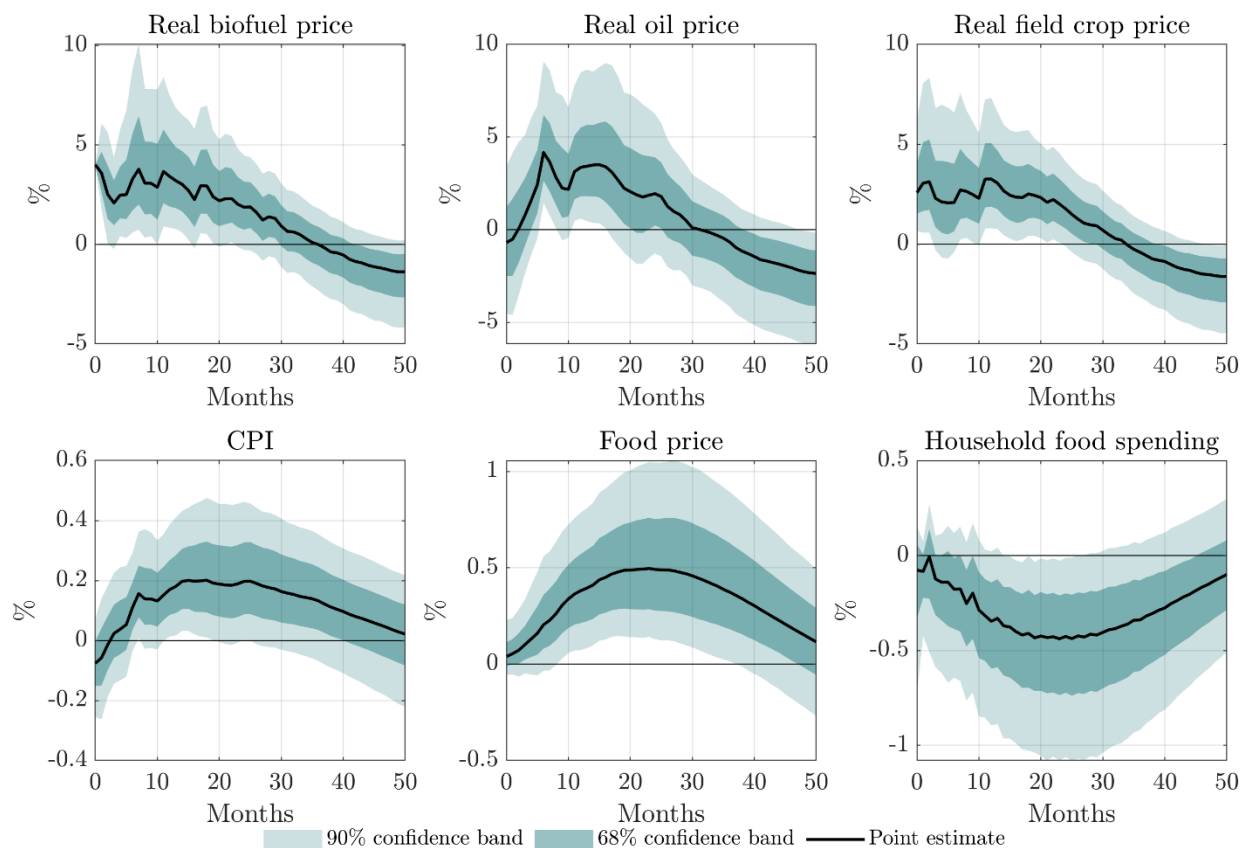
A shock that is normalized to raise biofuel prices by 4 percent increases oil prices by around 3.8 percent at month fourteen, suggesting that higher biofuel compliance costs transmit to petroleum markets through the blending relationship. In response to the shock, real field crop prices peak at around 3.6 percent after one year, more than twice the direct feedstock transmission of 1.2-1.6 percent calculated by a back-of-the-envelope calculation.⁶ This gap suggests that indirect channels amplify the initial demand shock: acreage competition between food and fuel crops reduces the supply of non-feedstock crops, while low short-run supply elasticity prevents production from expanding quickly enough to absorb the additional demand (Roberts and Schlenker, 2013).

Beyond the agricultural and energy sectors, the biofuel policy shock transmits to broader cost-of-living increases that bear directly on consumers. The shock raises the headline CPI by 0.22 percent after fifteen months. Food prices rise by 0.55 percent at their peak in month twenty-three, somewhat larger than the direct pass-through from field crop prices would predict: multiplying the peak field crop price response of 3.6 percent by the 12.7 percent average farm share of the food-at-home dollar over 2007—2023 implies an expected increase of 0.46 percent (ERS, 2026). The gap reflects cost channels beyond the farm commodity itself, since biofuel policy also raises energy and transportation costs that enter food processing and distribution. Household food spending reaches a trough of about 0.48 percent at month twenty-three, comparable to the food price increase, suggesting that consumption adjusts approximately one-for-one, although these responses are estimated with less precision.

This contraction in household food spending may reflect the broader inflationary effects of the shock. By raising energy, food, and overall prices simultaneously, tighter biofuel policy reduces household purchasing power and forces consumers to reallocate spending. Section 5.5 examines how these prices effect differ across income groups.

⁶ The direct effect is calculated as the biofuel price shock (4%) multiplied by the average feedstock usage share for corn and soybean oil (30%-40%), yielding an expected direct price transmission of approximately 1.2%-1.6%. The field crop response of 3.6% implies an amplification factor of around 2.5 relative to this benchmark.

Figure 5. Impulse response functions to a biofuel policy shock



Notes: The shock is normalized to a four percent increase in the real biofuel price (first variable, upper left). Impulse responses are plotted over a fifty-month horizon. Confidence intervals are derived from the moving-block bootstrap with 10,000 replications.

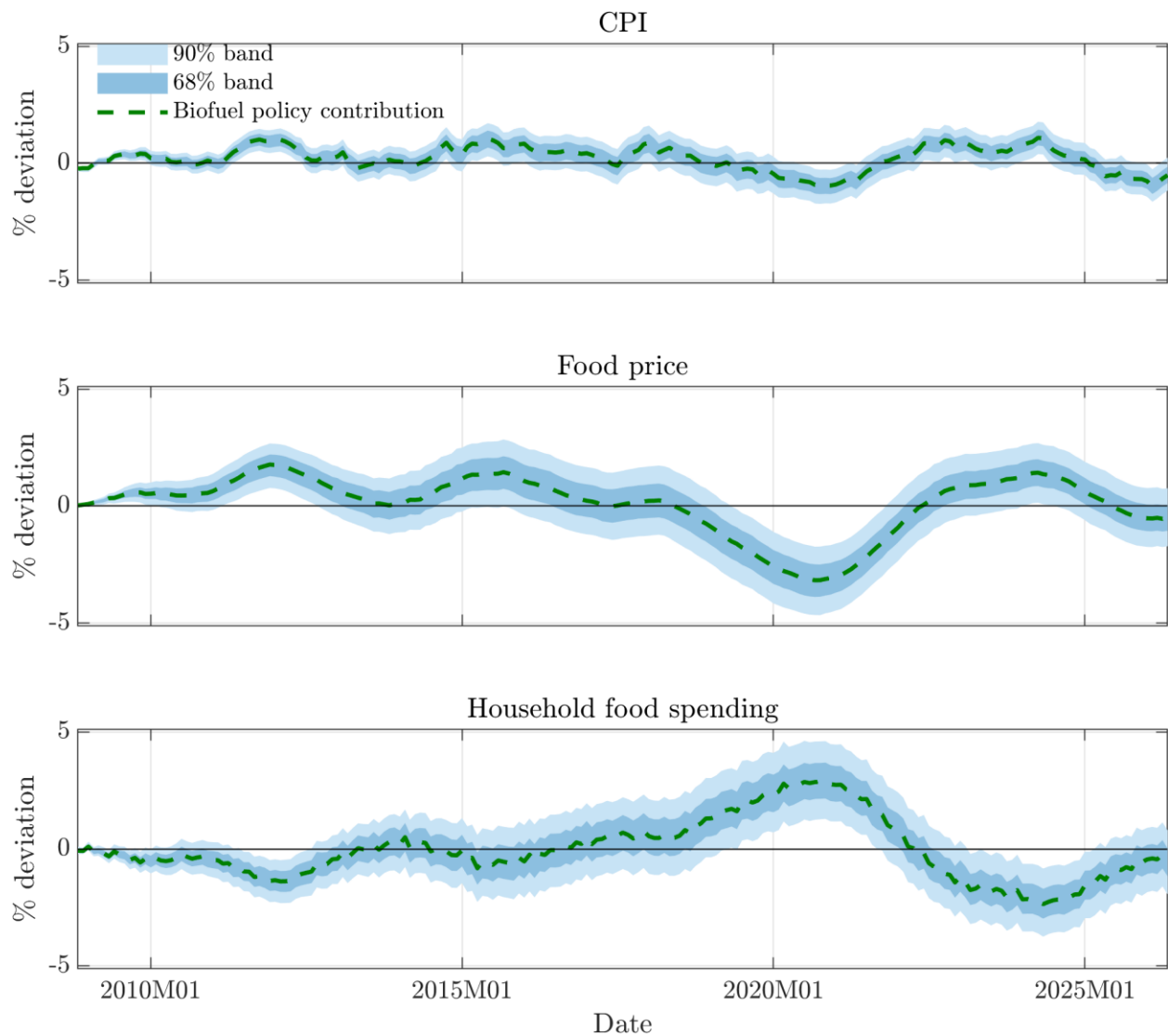
5.2 Contribution of biofuel policy shocks to downstream food markets

We next use historical decompositions to quantify how much these shocks contributed to observed price and expenditure movements. The contribution is obtained by propagating the identified shocks through the estimated VAR dynamics and cumulating their effects. Because we identify only one structural shock, the decomposition is partial. The remaining variation reflects all other shocks and initial conditions.

Figure 6 displays the contribution to three downstream series: the headline CPI, food prices, and household food spending. All series are in log levels, so contributions are expressed as percent deviations from the path that would have prevailed absent biofuel policy shocks. The contributions are economically meaningful and persistent. From 2019 through 2021, biofuel policy shocks held food prices below their counterfactual path, coinciding with an expansion of

small refinery exemptions that reduced aggregate compliance obligations and weakened demand for biofuel feedstocks. The sign reverses from mid-2022 through mid-2025: following the EPA's June 2023 Set Rule, which raised biomass-based diesel volumes relative to the proposed rule (EPA, 2023*b*), stronger compliance obligations increased feedstock demand, contributing to food prices above their counterfactual path, peaking at 1.4 percent in April 2024. The contribution to the headline CPI is smaller, peaking at 1.1 percent in the same month. Household food spending moves in the opposite direction from food prices, reflecting reduced real consumption when prices rise.

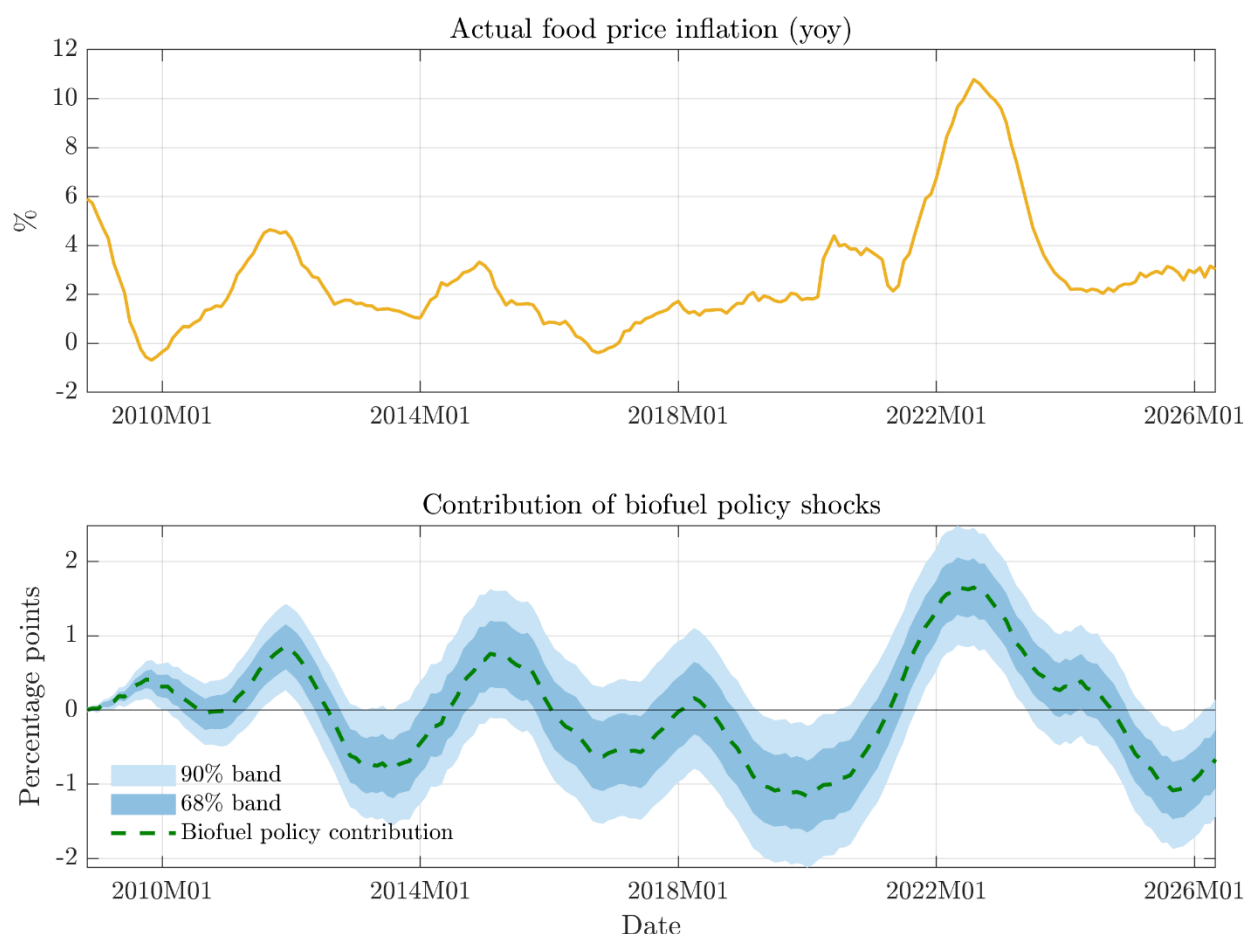
Figure 6. Contribution of biofuel policy shocks to downstream prices



Notes: The figure shows the cumulative historical contribution of biofuel policy shocks to four downstream price series from November 2008 to May 2026, expressed as percent deviations from the counterfactual path that would have prevailed absent these shocks.

Because policy debate centers on food price inflation rather than the price level, we repeat the exercise for the year-over-year food price inflation rate. Figure 7 plots realized inflation in the upper panel and the estimated contribution in the lower panel. The shocks contributed materially to the 2021-2023 surge, adding between 0.7 and 1.7 percentage points and remaining statistically significant throughout. At the August 2022 peak, food price inflation reached 10.8 percent; absent biofuel policy shocks, it would have been 9.1 percent. Biofuel policy thus accounted for roughly one-sixth of peak inflation, an economically meaningful contribution though not the dominant driver of an episode shaped primarily by historically large fiscal stimulus, supply-chain disruptions, and energy price shocks (Adjemian, Li, and Jo, 2026).

Figure 7. Contribution of biofuel policy shocks to food price inflation



Notes: The upper panel shows realized year-over-year food price inflation. The lower panel shows the estimated contribution of biofuel policy shocks, in percentage points. The sample spans November 2007 to May 2026.

5.3 Forecast error variance decomposition

Table 3 reports forecast-error variance decompositions. We extend the analysis to three supplementary variables not in the baseline model: soybean oil prices, food-away-from-home prices, and the U.S. exchange rate. Soybean oil is a major biofuel feedstock, food-away-from home prices capture the restaurant segment of food expenditures, and the exchange rate may amplify or dampen the domestic price effects through import-price pass-through.

Biofuel policy shocks account for 42 percent of real biofuel price variance, declining to 19 percent at four years, and a stable 14 to 18 percent of field crop price variance across horizons. Its contribution to soybean oil price variance peaks at 28 percent in the first year before easing to 17 percent in four years. Downstream, the shares are negligible on impact and rise with the horizon as cost pressures work through supply chains and retail pricing. Food-at-home price variance rises from zero to 28 percent in two years, while the food-away-from-home share climbs monotonically to 29 percent in four years, the most back-loaded of the series and consistent with slower repricing of restaurant menus. The headline CPI share is 22 to 23 percent in two and three years. Household food spending peaks near 10 percent at two to three years, and real oil prices and the exchange rate are barely explained at any horizon.

Table 3. Forecast error variance decomposition

Variable	0 mo.	12 mo.	24 mo.	36 mo.	48 mo.
Biofuel price	42.2	26.3	23.7	20.1	18.6
Oil price	3.4	7.7	8.6	7.7	7.7
Field crop price	16.4	14.1	18.2	16.3	15.9
Soybean oil price	20.2	28.0	23.2	17.9	16.8
CPI	18.3	12.1	21.9	23.1	21.4
Food-At-Home price	0.4	18.5	28.0	25.2	22.6
Food-Away-From-Home price	5.4	7.6	22.0	28.3	29.1
Household food spending	0.1	2.8	10.0	9.8	7.4
Exchange rate	0.0	8.3	7.0	5.8	6.1

Notes: Entries are the percent share of forecast-error variance attributable to biofuel policy shocks at horizons of 0, 12, 24, 36, and 48 months.

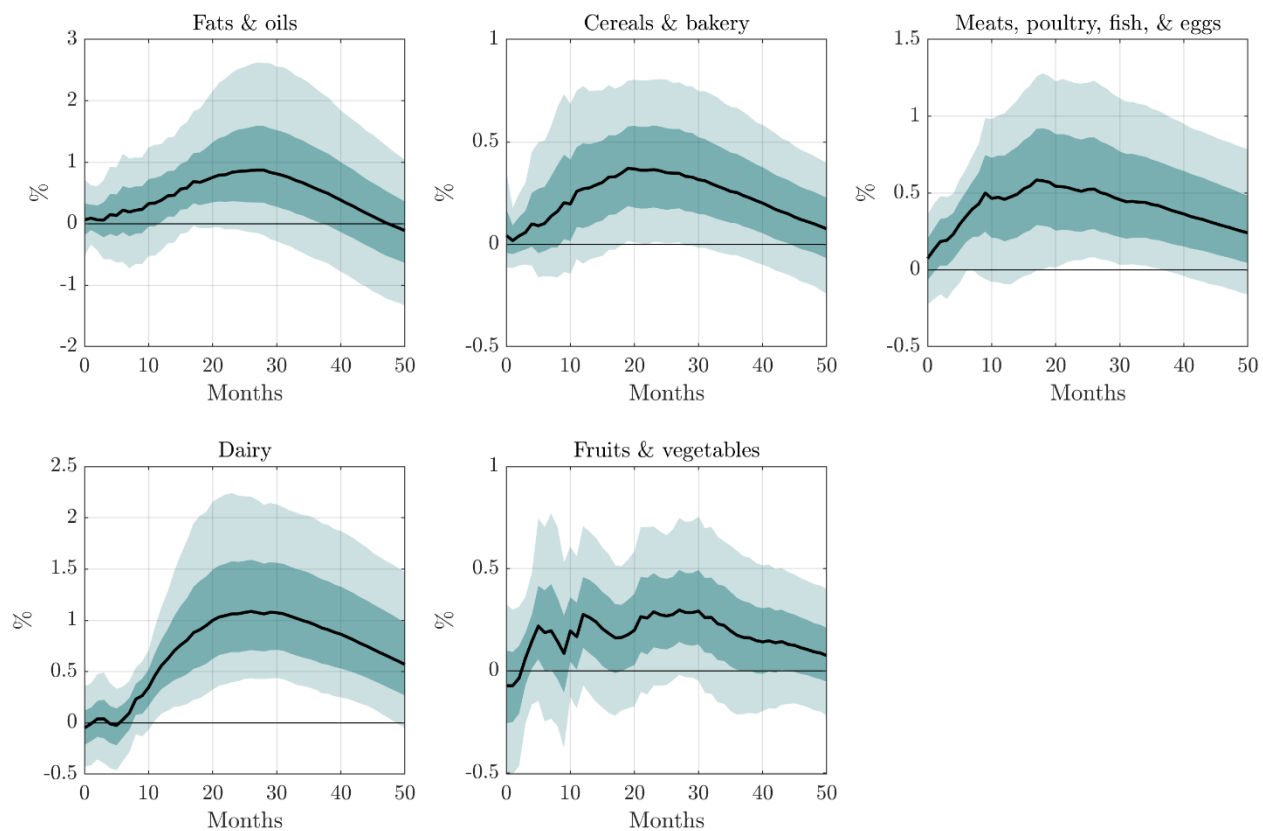
5.4 Heterogeneous price responses and broader inflationary effects

To identify where in the supply chain these effects originate and how they propagate, we analyze effects on granular food price components (CPI and PPI), livestock markets, retail energy prices, and inflation expectations. Once again following Känzig (2021), we augment the baseline SVAR with one variable at a time.

Figure 8 shows that food prices generally rise following the shock, but magnitudes range widely across categories. Fats and oils and dairy prices increase twice as much as other food subcategories, though through different channels. The fats and oils response reflects feedstock competition between food and fuel uses of soybean oil and corn, while the dairy response reflects higher livestock production costs as corn is diverted from animal feed markets (Schmit, Verteramo, and Tomek, 2009). Meats, poultry, fish, and eggs show smaller but positive responses.

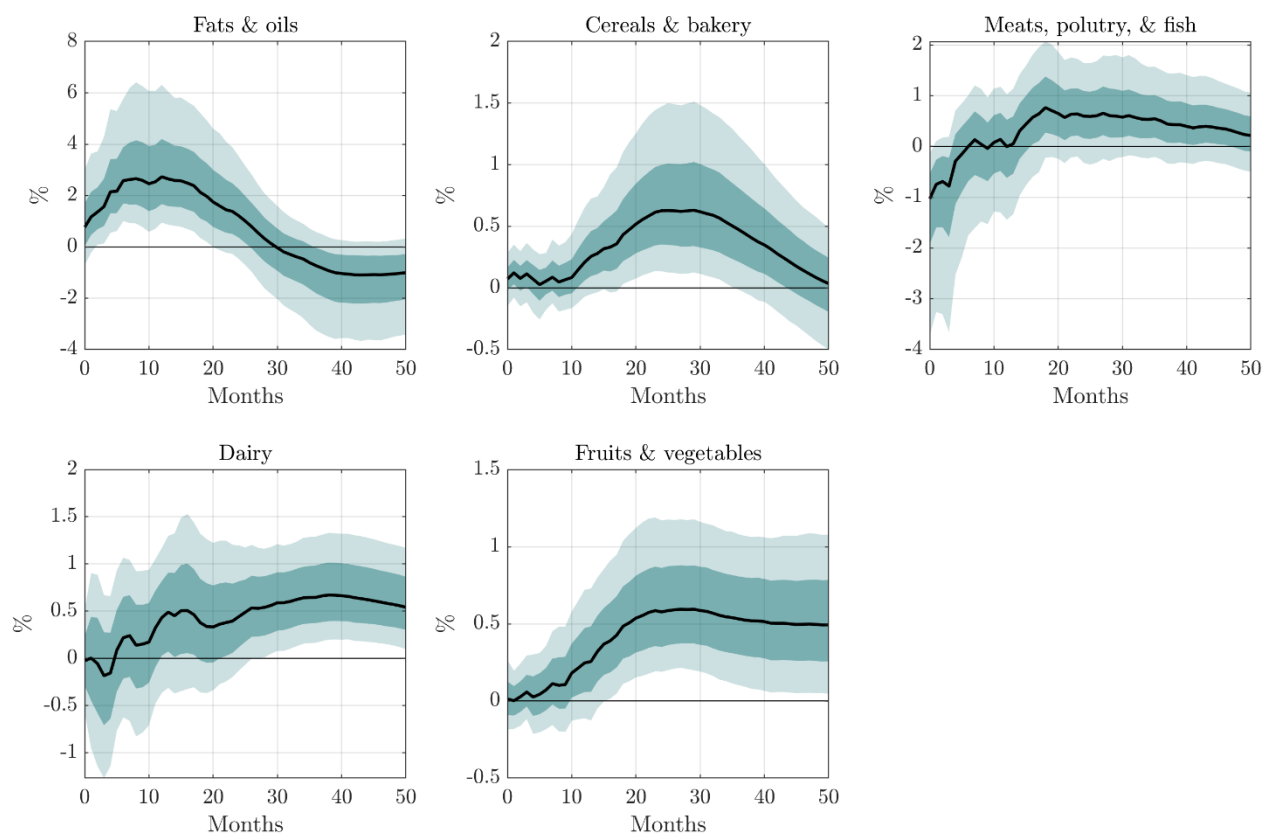
As shown in Figure 9, producer prices respond more strongly than their CPI counterparts, particularly for fats and oils. The PPI for fats and oils rises by about 2.7 percent at its peak after one year, more than twice the corresponding CPI response. This gap is anticipated because processors and retailers absorb part of the upstream cost pressure, so transmission weakens as the shock moves down the supply chain. Because biofuel policy affects livestock markets directly through the feed cost channel, we extend the analysis to cattle and beef markets. Figure 10 shows that the shock raises live cattle prices and beef and veal prices with a lag, while cattle placements respond little in the short run before declining at longer horizons.

Figure 8. Impulse responses of food price subcategories (CPI)



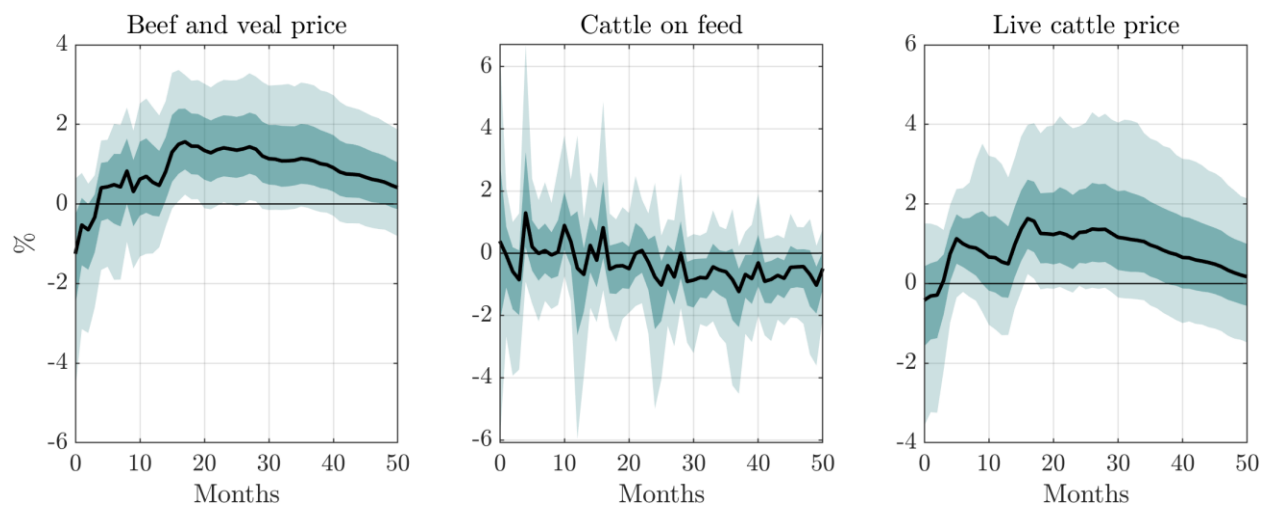
Notes: The shock is normalized to a 4 percent increase in the real biofuel price. Bands as in Figure 5. Impulse responses are estimated by augmenting the baseline VAR with each Consumer Price Index (CPI) food price subcategory one at a time. Data are from the Bureau of Labor Statistics.

Figure 9. Impulse responses of producer price index (PPI) food subcategories



Notes: The shock is normalized to a 4 percent increase in the real biofuel price. Bands as in Figure 5. Impulse responses are estimated by augmenting the baseline VAR with each Producer Price Index (PPI) food price subcategory one at a time. Data are from U.S. Bureau of Labor Statistics.

Figure 10. Impulse responses of livestock market

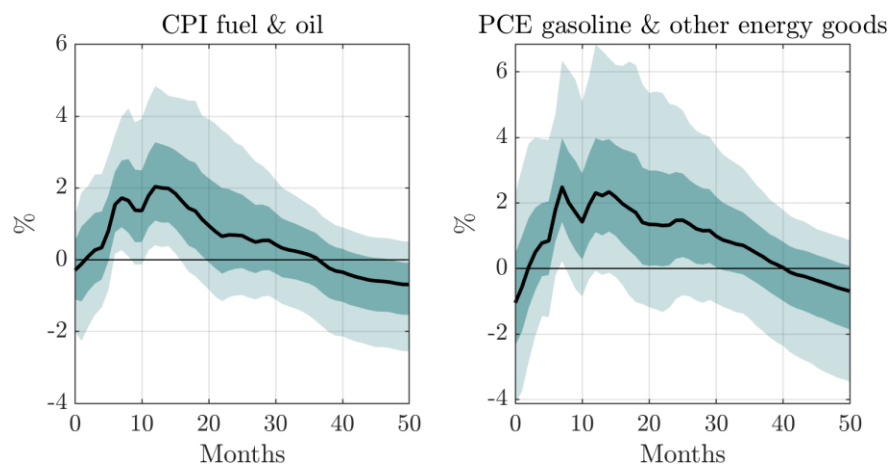


Notes: The shock is normalized to a 4 percent increase in the real biofuel price. Bands as in Figure 5. Impulse responses are estimated by augmenting the baseline VAR with PPI beef and veal price, cattle on feed, and live cattle

prices one at a time. Data are from U.S. Bureau of Labor Statistics, USDA National Agricultural Statistics Service, and Bloomberg terminal.

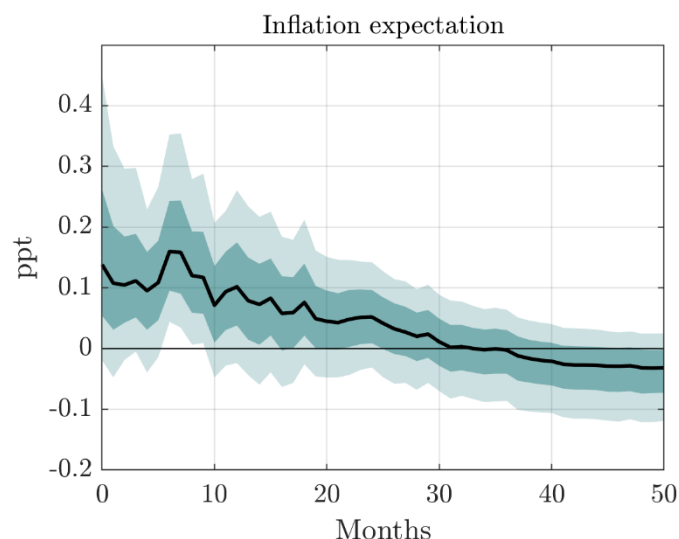
Figure 11 displays impulse responses of retail energy prices. CPI fuel and oil prices and PCE gasoline prices rise with a lag, indicating gradual and modest pass-through of upstream biofuel and oil price increases to retail energy prices (Lade and Bushnell, 2019). Figure 12 shows that consumer inflation expectations peak at 16 basis points six months after the shock. This suggests that persistent price increases in food and energy markets feed into forward-looking household beliefs about future inflation.

Figure 11. Impulse responses of energy prices



Notes: The shock is normalized to a 4 percent increase in the real biofuel price. Bands as in Figure 5. Impulse responses are estimated by augmenting the baseline VAR with each biofuel price one at a time. Data are from the U.S. Bureau of Labor Statistics and Bureau of Economic Analysis.

Figure 12. Impulse responses of inflation expectations



Notes: The shock is normalized to a 4 percent increase in the real biofuel price. Bands as in Figure 5. Impulse responses are estimated by augmenting the baseline VAR with inflation expectations. Inflation expectations represent the median expected price change over the next 12 months from the University of Michigan Surveys of Consumers. Data are from the University of Michigan: Inflation Expectation series, retrieved from FRED (accessed at: <https://fred.stlouisfed.org/series/MICH>).

5.5. Distributional consequences

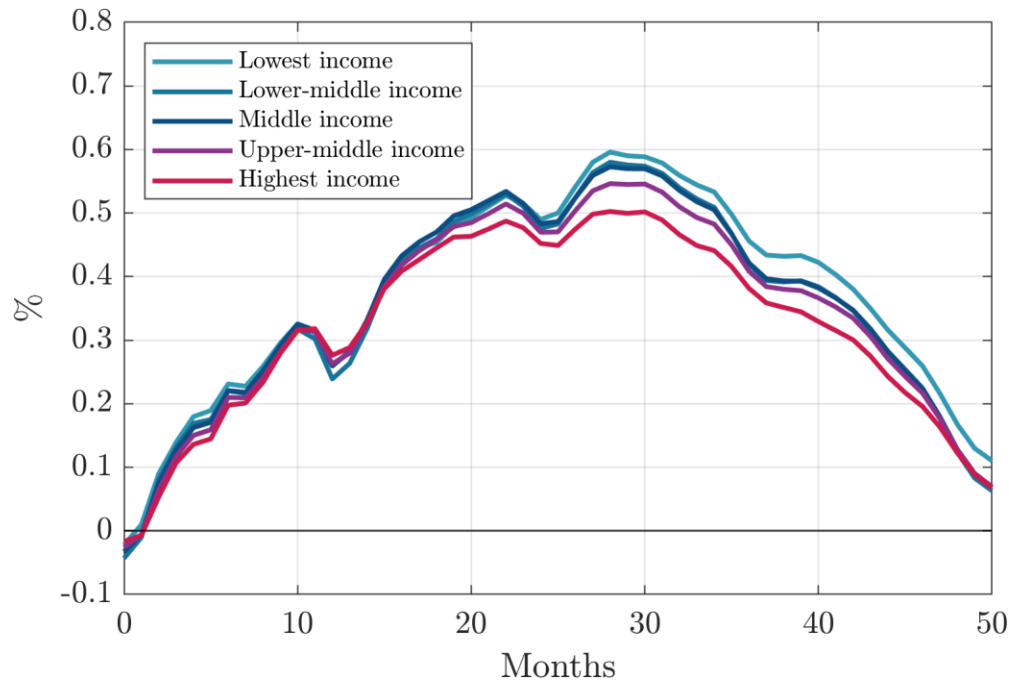
We now turn to how the inflationary burden of biofuel policy is distributed across households. Because food and electricity are basic necessities that constitute a larger share of budgets among lower-income households, a common price risk need not impose a common burden. Following Jaravel (2026), we construct a distributional CPI (D-CPI) for food and electricity by income quintile and compare impulse responses across groups.⁷ The D-CPI follows the construction of the official CPI but applies expenditure weights calculated separately for each income group. As in the previous section, we augment the baseline model with one D-CPI variable at a time.

As shown in Figure 13, the responses decline monotonically in income once the responses diverge after two years. The lowest income quintile's D-CPI peaks at 0.6 percent at month 28, compared with 0.5 percent for the highest quintile. These results indicate that biofuel policy shocks impose a proportionally larger inflationary burden on lower-income households,

⁷ The D-CPI measures inflation separately for each income group by constructing group-specific price indices using expenditure weights drawn from the Consumer Expenditure Survey (Jaravel, 2026). We isolate the food and electricity component by restricting the expenditure weights to those categories, following the replication code provided by Jaravel (2026). All indices are normalized to 100 in January 1983.

consistent with their greater expenditure shares on food and energy (Sinclair and Stewart, 2026). This burden may grow as U.S. biofuel policy expands through year-round E15 access, Sustainable Aviation Fuel (SAF) mandates, LCFS expansions, and clean fuel tax credit structures under the Inflation Reduction Act. Summary statistics and impulse responses with confidence intervals are reported in Appendix E.

Figure 13. Impulse responses of food and electricity distributional CPIs across income groups



Notes: The shock is normalized to a 4 percent increase in the real biofuel price. Point estimates from the impulse responses are plotted over a fifty-month horizon. Each response is estimated by augmenting the baseline VAR with one income-group distributional CPI at a time. We construct the food and electricity D-CPI based on Xavier Jaravel's replication code, published in his webpage (<https://www.xavierjaravel.com/dcpi>).

6. Robustness checks

This section assesses the sensitivity of the baseline results to instrument predictability, lag truncation, and alternative measurement of biofuel prices. The findings remain broadly consistent across specifications.

6.1 Instrument robustness to monetary policy

The informational sufficiency test indicates that the monetary policy block predicts the raw biofuel policy news series. As a robustness check, we re-estimate the baseline model using an

instrument orthogonalized against the federal funds rate. The impulse responses are nearly indistinguishable from the baseline (Appendix C).

6.2 Lag truncation bias: Local Projections

Impulse response reliability in finite-order VAR (p) models depends on lag length selection, and misspecification can introduce lag truncation bias that distorts estimated long-run dynamics (Montiel Olea et al., 2025). Local projections avoid this problem by estimating responses at each horizon, without imposing the cross-horizon restrictions embedded in a fitted VAR. We therefore re-estimate IRFs using local projections following Jordà (2005):

$$y_{t+h} = \alpha_h + \delta_h u_{1,t} + \theta'_h x_{t-1} + \xi_{h,t} \quad (10)$$

where y_{t+h} denotes the outcome variable at horizon h , $u_{1,t}$ represents the structural biofuel policy shock recovered from the baseline SVAR-IV, and x_{t-1} is a vector of baseline variables, and δ_h captures the response at horizon h .

We use the structural shock rather than the raw news series to address statistical power concerns in local projections (Nakamura and Steinsson, 2018). The two approaches yield broadly similar results, particularly at short horizons. At longer horizons, local projection estimates are somewhat more volatile and less precise, as each horizon is estimated independently without the smoothing imposed by VAR's dynamic structure, and tend to be larger than the baseline estimates (Appendix D).

6.3 Alternative measurement: ethanol price and shock normalization

As a sensitivity check, we first replace the real biofuel price with the ethanol futures price; the responses are comparable. We then estimate the model using a one-standard-deviation shock in place of the baseline normalization. The signs are consistent with the baseline and the responses are more precisely estimated (Appendix D).

7. Conclusions

In this study, we provide new empirical evidence on how U.S. biofuel policy transmits through agri-food supply chains to retail prices and consumer expenditures. We construct a biofuel policy news series based on regulatory announcements from the EPA and CARB, along with *Reuters*

reports of leaked information regarding Renewable Volume Obligations, and use it as an external instrument in a structural VAR framework. This approach allows us to trace the complete causal pathway from biofuel mandates through commodity markets to retail prices.

We find that tighter biofuel policy generates significant inflationary pressure across both upstream commodity and downstream consumer markets. Upstream, the shock raises biofuel prices, field crop prices, oil prices. Downstream, it raises consumer prices and food prices, while household food spending declines. The inflationary effects are concentrated in food categories most exposed to feedstock competition, particularly fats and oils and dairy. Producer food prices respond more strongly than CPI, indicating that upstream cost pressures are partially absorbed before reaching consumers. In livestock markets, higher feed costs raise live cattle and beef prices while reducing feedlot placements at longer horizons. Retail energy prices rise with a lag and consumer inflation expectations increase, suggesting that tightening biofuel policy shapes not only realized prices but also forward-looking household beliefs about future inflation.

Historical decompositions show that biofuel policy shocks contributed meaningfully to downstream prices in both directions, holding food prices below their counterfactual path during the small refinery exemption period and above it following the 2023 Set Rule. These shocks account for roughly 1.7 percentage points of the 10.8 percent peak in food price inflation in August 2022, about one-sixth of the surge. Forecast error variance decompositions confirm that the contribution to consumer food prices grows substantially over time, reaching around one-quarter at longer horizons, consistent with gradual pass-through through supply chains and retail pricing. The distributional analysis reveals that the inflationary consequences are regressive: lower-income households face larger price increases than higher-income households.

These findings speak directly to ongoing U.S. biofuel policy debates. The recent expansion of year-round E15 access, which allows higher ethanol blends to be sold across more states during summer months, raises feedstock demand in much the same way as the policy stringency shocks we identify. Our results suggest that such expansions, while supporting revenues for field-crop and biofuel producers, raise prices for consumers at the retail level, with the heaviest burden falling on lower-income consumers. Similarly, as Low Carbon Fuel Standards spread beyond

California to states such as Oregon, Washington, and New Mexico, the compliance cost pressures are likely to grow, with corresponding effects on downstream price transmission.

Several caveats qualify these conclusions. The news series captures policy information conveyed through regulatory announcements and press reports, and surprises transmitted outside this set are not reflected in the instrument. The baseline estimates rest on a partial invertibility condition, although the internal instruments and local projections exercise in Section 6 deliver consistent transmission patterns. Finally, the distributional analysis relies on income-group expenditure shares rather than household-level price experiences, so the regressivity estimates capture average differences across groups and not the full dispersion within them.

It should be noted that our analysis does not provide a comprehensive welfare evaluation of U.S. biofuel policy. What we do establish is that the costs of these mandates fall systematically on consumers, with the heaviest burden on lower-income households who are least able to absorb them. As policymakers evaluate future Renewable Volume Obligations, LCFS expansions, and clean fuel tax credit structures under the Inflation Reduction Act, a complete accounting of transmission pathways from regulatory announcements through commodity markets to household budgets is necessary for balancing the environmental objectives of renewable fuel policy against its economic and equity consequences.

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Online Appendix

Appendix A: U.S. biofuel policy announcements

Table A.1. U.S. biofuel policy announcements, 11/2007-5/2026

Trading day	Announcements	Source
11/27/2007	EPA published a Notice in the Federal Register announcing a renewable fuel standard for 2008 of 4.66%	EPA
2/8/2008	EPA Proposes Regulations for 2007 and Beyond Renewable Fuel Standard (RFS) Program	Reuters
11/17/2008	Renewable Fuel Standard Increased for 2009	EPA
5/5/2009	EPA Lays out a Plan for the Nation's Increase in Renewable fuels	EPA
3/26/2010	Regulation of Fuels and Fuel Additives: Changes to Renewable Fuel Standard Program (Final rule)	EPA
7/12/2010	EPA Proposes 2011 Renewable Fuel Standards	EPA
11/29/2010	EPA Finalizes 2011 Renewable Fuel Standards	EPA
6/21/2011	EPA Proposes 2012 Renewable Fuel Standards, 2013 Biomass-Based Diesel Volume	EPA
8/23/2011	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
10/28/2011	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
12/27/2011	EPA Finalizes 2012 Renewable Fuel Standards	EPA
2/2/2012	Southwest Iowa Renewable Energy, LLC to Pay \$10,150 for Risk Management Plan Violations at Council Bluffs Ethanol Plant	EPA
2/21/2012	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
4/2/2012	EPA to Allow 15 Percent Renewable Fuel in Gasoline/Agency approves first applications for registration of ethanol to make E15	EPA
6/25/2012	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
11/1/2012	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
1/2/2013	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
1/31/2013	EPA Proposes 2013 Renewable Fuel Standards	EPA
4/1/2013	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
8/1/2013	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
8/6/2013	EPA Finalizes 2013 Renewable Fuel Standards to Help Promote American Energy Independence, Reduce Carbon Pollution/EPA also announces steps to address concerns about the E10 blend wall	EPA
10/1/2013	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
11/15/2013	EPA Proposes 2014 Renewable Fuel Standards / Proposal Seeks Input to Address "E10 Blend Wall," Reaffirms Commitment to Biofuels	EPA
1/28/2014	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
4/8/2014	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
7/8/2014	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
10/20/2014	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
1/7/2015	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
3/17/2015	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
6/1/2015	EPA Proposes Increases in Renewable Fuel Levels	EPA

7/1/2015	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
9/24/2015	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
11/30/2015	EPA Finalizes Increases in Renewable Fuel Levels	EPA
12/14/2015	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
4/18/2016	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
5/18/2016	EPA Proposes Increase in Renewable Fuel Levels	EPA
7/29/2016	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
10/21/2016	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
11/23/2016	EPA Finalizes Increase in Renewable Fuel Volumes	EPA
1/24/2017	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
4/20/2017	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
7/5/2017	EPA Proposes RFS Volumes Reflective of Market Realities for 2018	EPA
8/3/2017	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
11/1/2017	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
12/1/2017	U.S. EPA ups biofuel targets slightly, draws scorn from refiners	Reuters
2/1/2018	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
4/26/2018	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
6/26/2018	Proposed Volume Standards for 2019 and the Biomass-Based Diesel Volume for 2020	EPA
7/11/2018	EPA scraps detailed plan to force U.S. refiners to blend more biofuels	Reuters
8/1/2018	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
9/20/2018	EPA details broad expansion of biofuel waiver program	Reuters
11/1/2018	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
11/30/2018	EPA Finalizes RFS Volumes for 2019 and Biomass Based Diesel Volumes for 2020	EPA
2/1/2019	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
3/12/2019	EPA Proposes Rule to Allow E15 Waiver and to Improve RIN Market Transparency	EPA
5/1/2019	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
5/31/2019	EPA Delivers on President Trump's Promise to Allow Year-Round Sale of E15 Gasoline and Improve Transparency in Renewable Fuel Markets	EPA
7/5/2019	EPA Proposes Renewable Volume Obligations for 2020 On Track to Meet Statutory Deadline	EPA
8/1/2019	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
11/1/2019	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
12/19/2019	EPA Fulfills Another Trump Administration Promise: Finalizes RFS Volumes for 2020 and Biomass Based Diesel Volumes for 2021	EPA
2/3/2020	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
5/1/2020	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
8/3/2020	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
9/9/2020	Exclusive: Trump administration to deny pending retroactive U.S. biofuel waivers — sources	Reuters
11/2/2020	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
1/12/2021	EXCLUSIVE Trump administration expected to grant biofuel waivers for some oil refiners - sources	Reuters
1/15/2021	Notice of Proposed Rulemaking for E15 Labeling	EPA
2/1/2021	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
5/3/2021	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
8/2/2021	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB

9/22/2021	EXCLUSIVE: Biden administration mulls big cuts to biofuel mandates in win for oil industry — document	Reuters
11/1/2021	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
12/7/2021	Proposed Volume Standards for 2020, 2021, and 2022	EPA
1/13/2022	Exclusive: Biden weighing cuts to 2022 ethanol blending mandate proposal	Reuters
2/1/2022	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
5/2/2022	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
6/1/2022	EXCLUSIVE: Biden administration likely to retroactively raise ethanol blending volumes for 2021, sources say	Reuters
8/1/2022	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
11/1/2022	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
12/1/2022	EPA Takes Next Steps in Renewable Fuel Standard Program for 2023–25	EPA
2/1/2023	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
3/1/2023	EPA proposes expanded sales of higher ethanol blend gasoline in US Midwest	Reuters
4/19/2023	Biden administration sees year-round sales of higher ethanol fuel by 2024	Reuters
5/1/2023	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
6/7/2023	EPA to remove proposed EV volumes from biofuel blending rule	Reuters
6/21/2023	Final Renewable Fuels Standards for 2023, 2024, and 2025	EPA
8/1/2023	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
11/1/2023	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
2/1/2024	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
2/21/2024	Exclusive: Biden administration to approve E15 gasoline expansion starting in 2025, sources say	Reuters
4/8/2024	Exclusive: Biden administration's initial SAF subsidy model to raise climate hurdle for ethanol	Reuters
4/19/2024	Exclusive: US EPA to announce temporary expanded sales of higher-ethanol gasoline blend — sources	Reuters
5/1/2024	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
8/1/2024	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
11/1/2024	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
1/13/2025	Exclusive: US to release clean fuel tax credit model that limits ethanol producers' credit access, sources say	Reuters
2/21/2025	Ahead of Summer Driving Season, EPA Allows Expanded E15 Access to Midwest States Year-Round	EPA
3/3/2025	US EPA will delay Midwest ethanol expansion in South Dakota, Ohio	Reuters
4/2/2025	US EPA's upcoming proposal on biofuel blending mandates to cover two years, sources say	Reuters
5/1/2025	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
5/15/2025	US EPA to rule on backlog of small refinery biofuel exemption requests, Zeldin says	Reuters
6/13/2025	US EPA set to propose biomass-based diesel quotas below industry expectations, sources say	Reuters
8/1/2025	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
8/22/2025	EPA Announces Action on Small Refinery Exemptions, Continues Work to Get Renewable Fuel Standard Program Back on Track	EPA
9/10/2025	Exclusive: US biofuel plan would reallocate half or less of waived blending quotas, sources say	Reuters

9/16/2025	EPA Proposes Renewable Fuel Standards Supplemental Rule, Addresses Small Refinery Exemptions	EPA
11/3/2025	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
11/20/2025	Trump administration may delay biofuel import credit cuts as refiners balk	Reuters
12/15/2025	Trump administration unlikely to finalize 2026 biofuel quotas this year, sources say	Reuters
2/2/2026	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB
3/27/2026	EPA Finalizes Historic New Renewable Fuel Standards to Strengthen American Energy Security, Support Rural Economies	EPA
5/1/2026	Low Carbon Fuel Standard (LCFS) quarterly reports	CARB

Notes: The table lists all 113 biofuel policy announcements used to construct the biofuel policy news series. CARB reports are released at 3 pm Pacific Time, after market close, so the assigned trading day is the following business day. Announcement dates therefore span November 2007 to April 2026, while assigned trading days extend to May 2026. EPA = Environmental Protection Agency; CARB = California Air Resources Board. Reuters announcements report on or anticipate EPA Renewable Volume Obligation decisions, and CARB announcements are quarterly compliance reports on Low Carbon Fuel Standard (LCFS) credits and deficits.

Appendix B: Data description and source

Table B. Data description and source for selected variables

Variable (Unit)	Description	Source
A. Instrument		
Biofuel policy news (%)	First principal component of daily changes in ethanol and soybean oil futures prices (nearby and first-deferred contract maturities) around regulatory announcements from EPA, CARB, and Reuters, weighted by monthly production shares	Bloomberg terminal, EIA, and authors' calculations
B. Baseline		
Real biofuel price	Production-weighted composite of the Chicago ethanol (Platts) nearby futures contract price and the OPIS wholesale biodiesel price by OPIS, aggregated using monthly production shares and deflated by CPI	Bloomberg terminal; Center for Agricultural and Rural Development (CARD), Iowa State University
Real oil price (index, 2017=100)	Global Brent crude spot price, deflated by CPI	FRED; BLS
Real field crop price (index, 2017=100)	Production-weighted average of nearby corn and soybean futures prices, deflated by CPI	Bloomberg terminal; USDA NASS; BLS

CPI (index, 1982-84=100)	Headline Consumer Price Index for all urban consumers (CPI-U), all items	BLS
Food-at-home CPI (index, 1982-84=100)	Consumer Price Index for food at home in U.S. city average (grocery store prices)	BLS
RPCE for food (Billions of Chained 2017 Dollars)	Real Personal Consumption Expenditures price: food, deflated by CPI	BEA
C. Wider Effects		
<i>Consumer food price indices (CPI)</i>		
Fats and oils CPI (index, 1982-84=100)	Consumer Price Index for fats and oils in U.S. city average	BLS
Cereals and bakery CPI (index, 1982-84=100)	Consumer Price Index for cereals and bakery products in U.S. city average	BLS
Meats, poultry, fish, and eggs CPI (index, 1982-84=100)	Consumer Price Index for meats, poultry, fish, and eggs in U.S. city average	BLS
Dairy CPI (index, 1982-84=100)	Consumer Price Index for dairy and related products in U.S. city average	BLS
Fruits and vegetables CPI (index, 1982-84=100)	Consumer Price Index for fruits and vegetables in U.S. city average	BLS
<i>Producer food price indices (PPI)</i>		
Fats and oils PPI (index, 1981=100)	Producer price index for fats, oils refining and blending: shortening and cooking oils	BLS
Cereal and bakery PPI (index, 1982=100)	Producer price index for cereal and bakery products	BLS
Meats, poultry, fish, and egg PPI (index, 1982=100)	Producer price index for meats, poultry, and fish	BLS
Dairy PPI (index, 1982=100)	Producer price index for dairy products	BLS
Fruits and vegetables PPI (index, 1982=100)	Producer price index for processed fruits and vegetables	BLS
<i>Livestock markets</i>		
Beef and veal PPI (index, 1982=100)	Producer price index for beef and veal products (fresh or frozen)	BLS
Cattle on feed (head)	Monthly placements of cattle into feedlots with 1,000+ head capacity, measured in head	USDA NASS Cattle on Feed
Live cattle price (cents/lb)	Chicago Mercantile Exchange (CME) nearby live cattle futures contract price	Bloomberg terminal
<i>Energy prices</i>		
CPI fuel and oil (index, 1982-84=100)	Consumer Price Index for fuel oil and other fuels	BLS

PCE gasoline (index, 2017=100)	Personal Consumption Expenditures price index for gasoline and other energy goods	BEA
Consumer inflation expectations (%)	University of Michigan Survey of Consumers: median expected price change over the next 12 months	University of Michigan
D. Distributional Effects		
<i>distributional CPI</i>		
Food-and-electricity distributional CPI	Distributional CPI for the combined food (BLS 'Food,' SAF1) and electricity (SEHF01) basket, across income quintile, constructed following Jaravel (2026) by reweighting BLS item-strata price relatives with group-specific expenditure shares from the CEX	BLS; CEX; Jaravel (2026); authors' calculations
E. Other Variables		
Exchange rate (index, 2020=100)	U.S. real narrow effective exchange rate	BIS
Ethanol production (Million Gallons)	U.S. fuel ethanol production	EIA
Real ethanol price (USD/gallon)	Chicago ethanol (Platts) nearby futures price, deflated by CPI	Bloomberg terminal
Biodiesel production (Million Gallons)	U.S. biodiesel production	EIA
Food-away-from-home CPI (index, 1982-84=100)	Consumer Price Index for food away from home in U.S. city average (restaurant prices)	BLS

Notes: All price variables are transformed to natural logarithms in the empirical analysis. Ethanol, oil, and field crop prices are deflated by the headline CPI. Sample period is November 2007 to February 2025. BLS = Bureau of Labor Statistics; BEA = Bureau of Economic Analysis; FRED = Federal Reserve Economic Data; CEX = Consumer Expenditure Survey; CARB = California Air Resources Board; EPA = Environmental Protection Agency; USDA = U.S. Department of Agriculture; BIS = Bank for International Settlements.

Appendix C: Informational-sufficiency tests and monetary-policy-orthogonalized instrument

C.1 Informational-sufficiency test results

We report the full results of the informational-sufficiency test introduced in Section 3.4. Proxy-SVAR identification requires that the instrument be correlated with the biofuel policy shock but uncorrelated with other structural shocks. A necessary condition, formalized by Mori and Peersman (2026), is that the instrument cannot be predictable from information available to agents at the time. If lagged macroeconomic or financial variables forecast the identified shock, the instrument is capturing anticipated or confounding movements rather than genuine policy news. Tables C.1 and C.2 present Granger-predictability p -values by information block and by

individual series, for the full sample and for a pre-COVID subsample ending in December 2019. Each cell tests whether twelve monthly lags of the relevant information predict one of three outcome objects: the raw news series, the reduced-form residual of the real biofuel price equation, or the identified structural shock. The *p (orth.)* column, which orthogonalizes the outcome on the baseline VAR variables, carries the verdict; large values indicate sufficiency.

Only the monetary policy block predicts the raw news series (*p*-value: 0.013; Panel A), and this predictability disappears after conditioning on the baseline VAR lags (orthogonalized *p* = 0.505). Table C.2 locates the raw result in the federal funds rate and the monetary policy shock of Bauer and Swanson (2023), neither of which survives orthogonalization. The remaining rejections are isolated and marginal: the global supply chain pressure index for the reduced-form residual in the pre-COVID subsample, and nonfinancial leverage and geopolitical risk for the structural shock. Because the monetary policy block is the only one predicting the raw series, we verify robustness by re-estimating with an orthogonalized instrument in Section C.2.

Table C.1. Informational-sufficiency tests for the biofuel-policy instrument (block-joint)

Information block	Full sample		Pre-COVID ($\leq 2019M12$)	
	<i>p (raw)</i>	<i>p (orth.)</i>	<i>p (raw)</i>	<i>p (orth.)</i>
Panel A. Instrument (surprise proxy)				
Financial / credit	0.696	0.905	—	—
Policy uncertainty	0.950	1.000	0.829	0.713
Commodity / global	0.293	0.949	0.644	0.953
Monetary policy	0.013	0.505	0.999	0.999
Panel B. Reduced-form residual				
Financial / credit	0.161	0.187	—	—
Policy uncertainty	0.396	0.440	0.188	0.186
Commodity / global	0.946	0.951	0.707	0.762
Monetary policy	0.907	0.905	0.782	0.768
Panel C. Structural shock				
Financial / credit	0.487	0.505	—	—
Policy uncertainty	0.279	0.288	0.041	0.043
Commodity / global	0.996	0.995	0.575	0.589
Monetary policy	0.981	0.979	0.661	0.656

Notes: Each cell reports the *p*-value of a Granger F-test that twelve monthly lags of the information block jointly predict the outcome object, controlling for a constant and twelve own lags. *p (raw)* uses the outcome as measured; *p (orth.)* uses the outcome after orthogonalizing it on 12 lags of the baseline VAR variables (the Mori-Peersman informational-sufficiency criterion). Small *p*-values indicate predictability, i.e., a potential sufficiency violation; large *p (orth.)* values indicate the instrument is sufficient against that block. “—” denote tests not computable on the pre-

COVID subsample (the nine-series financial block exhausts the available degrees of freedom at 12 lags; see the per-series table). Sample size ranges 168-199 (full) and equals 122 (pre-COVID). Series definitions and sources are given in the notes to Table C.2.

Table C.2. Informational-sufficiency tests for the biofuel-policy instrument (per series)

Series	Full sample		Pre-COVID ($\leq 2019M12$)	
	<i>p</i> (raw)	<i>p</i> (orth.)	<i>p</i> (raw)	<i>p</i> (orth.)
Panel A. Instrument (surprise proxy)				
<i>Financial / credit</i>				
Financial uncertainty (LMN)	0.826	0.997	0.914	0.949
Excess bond premium	0.720	0.926	0.916	0.711
Gilchrist–Zakrajšek spread	0.781	1.000	0.998	0.964
NFCI	0.405	0.811	0.924	0.970
Adjusted NFCI	0.275	0.930	0.911	0.928
NFCI: risk	0.302	0.858	0.951	0.985
NFCI: credit	0.573	0.933	0.981	0.978
NFCI: leverage	0.644	0.961	0.289	0.201
Nonfinancial leverage	0.455	0.979	0.707	0.937
<i>Policy uncertainty</i>				
Geopolitical risk (GPR)	0.812	1.000	0.723	0.797
Trade policy uncertainty (TPU)	0.896	0.987	0.777	0.863
Economic policy uncertainty (EPU)	0.684	0.883	0.623	0.227
<i>Commodity / global</i>				
Supply-chain pressure (GSCPI)	0.165	0.791	0.318	0.751
Oil supply news shock	0.509	0.892	0.935	0.891
Agricultural supply news shock	0.362	0.789	0.843	0.736
<i>Monetary policy</i>				
Federal funds rate	0.090	0.680	0.985	0.994
Monetary policy shock (BRW)	0.464	0.888	0.904	0.905
Monetary policy shock (BS)	0.010	0.958	0.915	0.985
Panel B. Reduced-form residual				
<i>Financial / credit</i>				
Financial uncertainty (LMN)	0.352	0.433	0.235	0.246
Excess bond premium	0.813	0.811	0.252	0.291
Gilchrist–Zakrajšek spread	0.995	0.995	0.385	0.422
NFCI	0.990	0.990	0.559	0.589
Adjusted NFCI	0.988	0.984	0.530	0.564
NFCI: risk	0.984	0.984	0.652	0.682
NFCI: credit	0.957	0.962	0.389	0.441
NFCI: leverage	0.622	0.614	0.479	0.501
Nonfinancial leverage	0.625	0.700	0.748	0.777
<i>Policy uncertainty</i>				
Geopolitical risk (GPR)	0.344	0.359	0.070	0.078
Trade policy uncertainty (TPU)	0.287	0.292	0.985	0.986

Series	Full sample		Pre-COVID ($\leq 2019M12$)	
	<i>p (raw)</i>	<i>p (orth.)</i>	<i>p (raw)</i>	<i>p (orth.)</i>
Economic policy uncertainty (EPU)	0.242	0.282	0.326	0.340
<i>Commodity / global</i>				
Supply-chain pressure (GSCPI)	0.551	0.543	0.076	0.088
Oil supply news shock	0.970	0.967	0.969	0.973
Agricultural supply news shock	0.846	0.865	0.819	0.832
<i>Monetary policy</i>				
Federal funds rate	0.999	0.999	0.809	0.815
Monetary policy shock (BRW)	0.823	0.808	0.746	0.744
Monetary policy shock (BS)	0.412	0.437	0.178	0.192
Panel C. Structural shock				
<i>Financial / credit</i>				
Financial uncertainty (LMN)	0.213	0.228	0.764	0.773
Excess bond premium	0.966	0.968	0.915	0.916
Gilchrist–Zakrajšek spread	0.728	0.724	0.175	0.182
NFCI	0.677	0.667	0.639	0.635
Adjusted NFCI	0.686	0.676	0.447	0.440
NFCI: risk	0.755	0.749	0.465	0.463
NFCI: credit	0.611	0.603	0.747	0.745
NFCI: leverage	0.429	0.430	0.676	0.682
Nonfinancial leverage	0.083	0.092	0.598	0.611
<i>Policy uncertainty</i>				
Geopolitical risk (GPR)	0.339	0.343	0.055	0.058
Trade policy uncertainty (TPU)	0.420	0.412	0.723	0.726
Economic policy uncertainty (EPU)	0.273	0.284	0.106	0.105
<i>Commodity / global</i>				
Supply-chain pressure (GSCPI)	0.850	0.852	0.735	0.739
Oil supply news shock	0.988	0.987	0.801	0.803
Agricultural supply news shock	0.811	0.801	0.480	0.490
<i>Monetary policy</i>				
Federal funds rate	0.998	0.998	0.479	0.491
Monetary policy shock (BRW)	0.550	0.547	0.691	0.681
Monetary policy shock (BS)	0.926	0.927	0.509	0.514

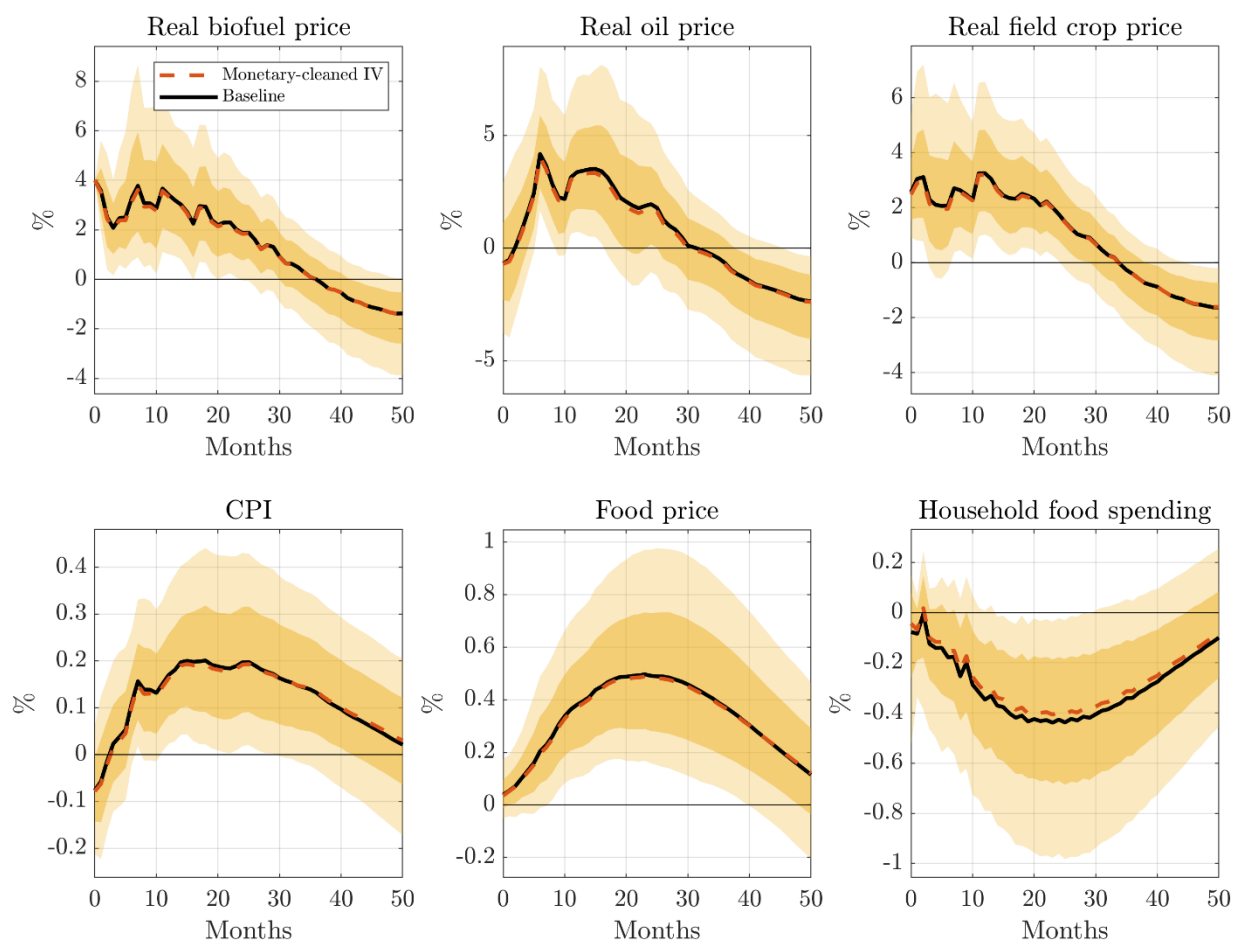
Series definitions and sources: Financial / credit: financial uncertainty (LMN), the one-month-ahead financial-uncertainty index of Ludvigson, Ma, and Ng (2021); excess bond premium (EBP) and the Gilchrist–Zakrajšek (GZ) spread, Gilchrist and Zakrajšek (2012), updated by the Federal Reserve Board; the National Financial Conditions Index (NFCI), the Adjusted NFCI (ANFCI), and its risk, credit, leverage, and nonfinancial-leverage subindexes, Federal Reserve Bank of Chicago. Policy uncertainty: Geopolitical Risk (GPR), Caldara and Iacoviello (2022); Trade Policy Uncertainty (TPU), Caldara et al. (2020); Economic Policy Uncertainty (EPU), Baker, Bloom, and Davis (2016). Commodity / global: Global Supply Chain Pressure Index (GSCPI), Federal Reserve Bank of New York; Oil supply news shock, Känzig (2021); Agricultural supply news shock, Adjemian and Jo (2025). Monetary policy: Federal Funds Effective Rate (FFR), Board of Governors of the Federal Reserve System; monetary policy shock (BRW), Bu, Rogers, and Wu (2021); monetary policy shock (BS), Bauer and Swanson (2023).

Notes: Per-series counterpart of Table C.1; each row tests a single series (constant, twelve own lags of the outcome, and twelve lags of the series). p (*orth.*) orthogonalizes the outcome on twelve lags of the baseline VAR variables; small values indicate predictability. Panels report the three outcome objects: the news series (Panel A), the reduced-form residual (Panel B), and the structural shock (Panel C). N = 199 (full) / 122 (pre-COVID) except BRW (168), BS (171 full), oil supply news shock (195), and agricultural supply news shock (183).

C.2 Monetary-Policy-orthogonalized instrument

Although the predictability does not survive conditioning on the baseline model, it raises the possibility that part of the news series reflects anticipated policy conditions. As a robustness check, we orthogonalize the instrument against the federal funds rate and retain the residual, rescaled to the mean and standard deviation of the original series. The heteroskedasticity-robust first-stage F-statistic is 20.3. Figure C.1 compares the impulse responses with the baseline. The impulse responses are close throughout and match the baseline in both sign and shape.

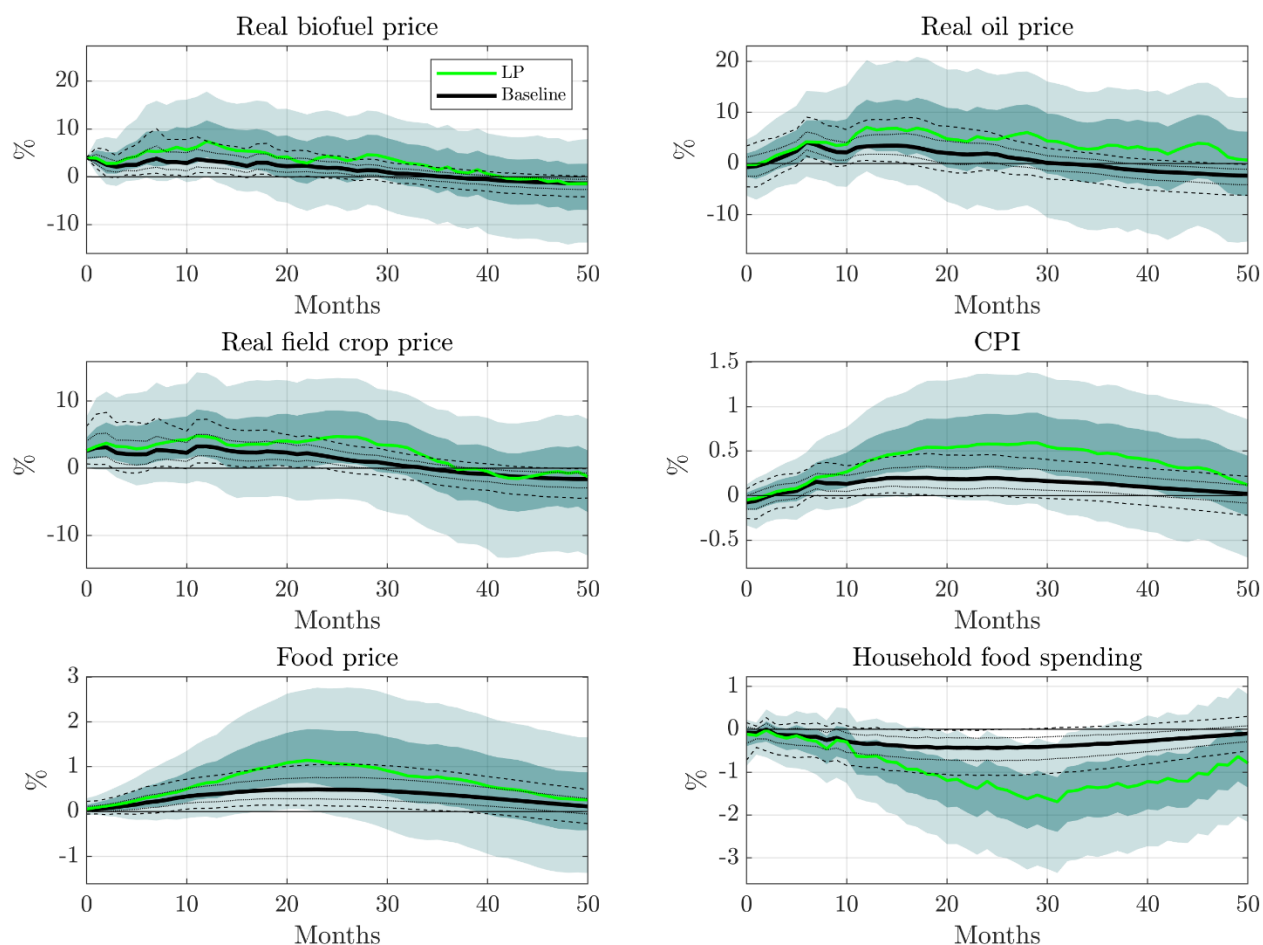
Figure C.1. Impulse responses to a biofuel policy shock: monetary policy-orthogonalized instrument versus baseline



Notes: The figure compares impulse responses identified with the monetary-orthogonalized instrument (dashed line, with shaded 68 and 90 percent moving-block bootstrap confidence bands) to the baseline responses estimated with the original instrument (solid black line). The shock is normalized to a four percent increase in the real biofuel price on impact. The policy-cleaned instrument is the residual from a regression of the original instrument on six lags of the effective federal funds rate, rescaled to the original mean and standard deviation; its heteroskedasticity-robust first stage F-statistic is 20.3.

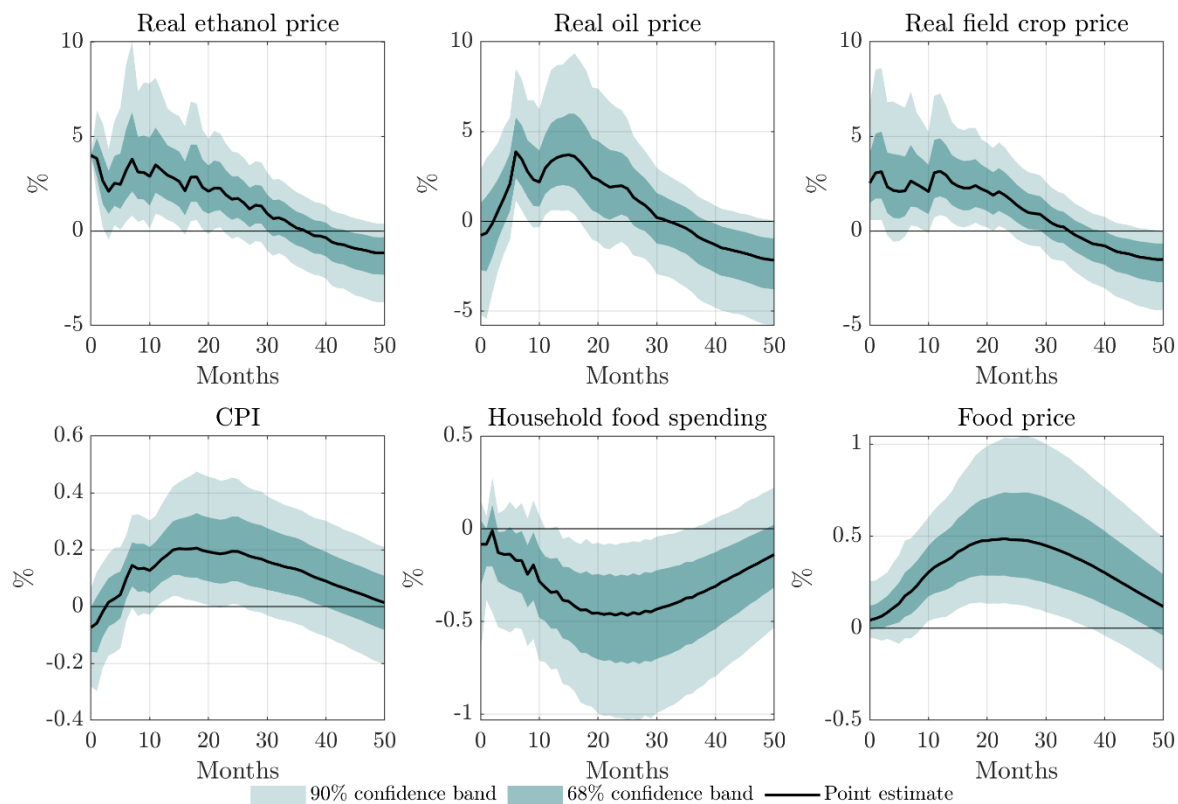
Appendix D: Sensitivity and robustness checks

Figure D.1. Impulse responses to a biofuel policy shock using local projections



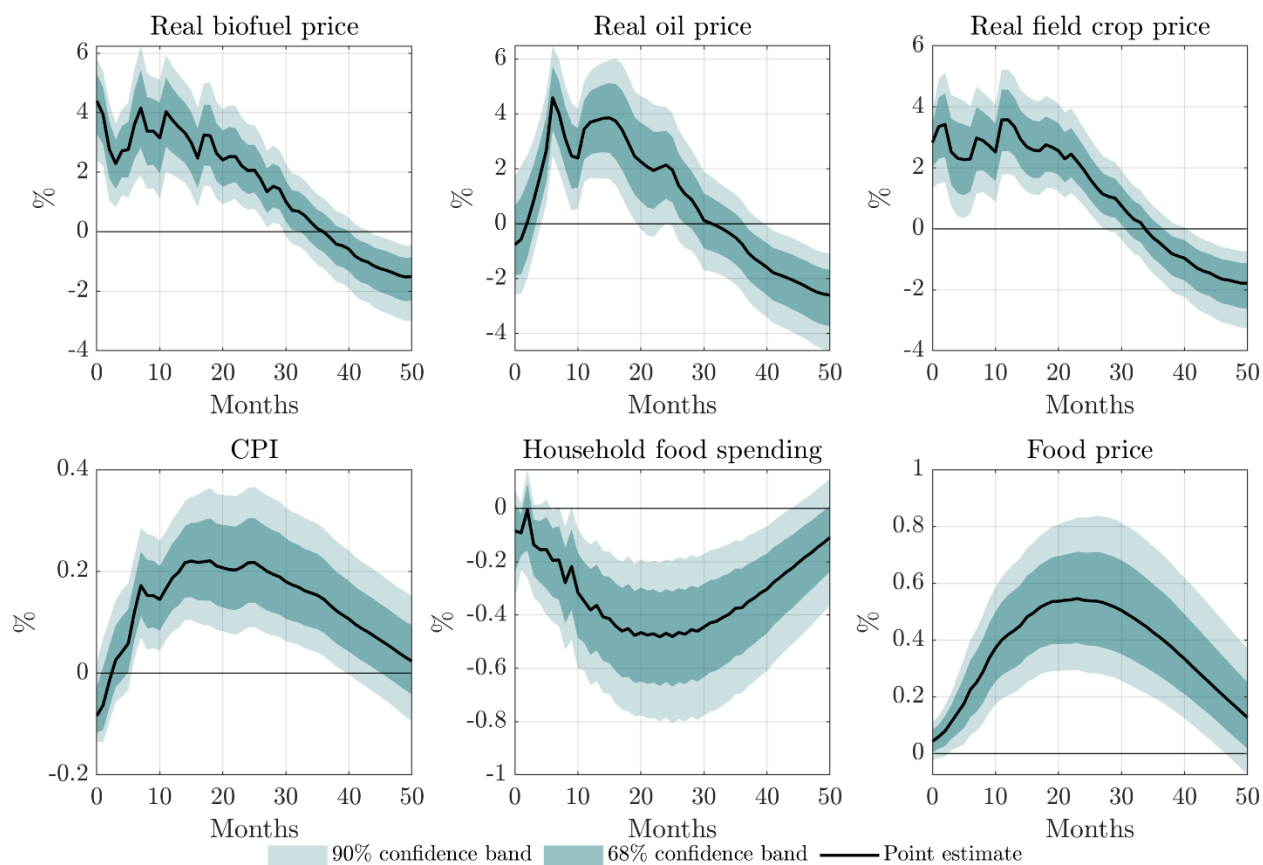
Notes: Impulse responses are estimated using both the local projections and external instruments (SVAR-IV) approach. Responses are normalized to raise the real ethanol price by 2 percent. The green solid line indicates the point estimate from the internal instruments approach, and the black solid line represents the estimate from the external instruments approach. Bands as in Figure 5.

Figure D.2. Impulse responses with ethanol price



Notes: This figure re-estimates the baseline model with the real ethanol price replacing the real biofuel price as the first variable. The shock is normalized to a two percent increase in the real ethanol price (first variable, upper left). Impulse responses are plotted over a fifty-month horizon. The heteroskedasticity-robust first-stage F-statistic is 10.19. Bands as in Figure 5.

Figure D.3. Impulse responses without normalization



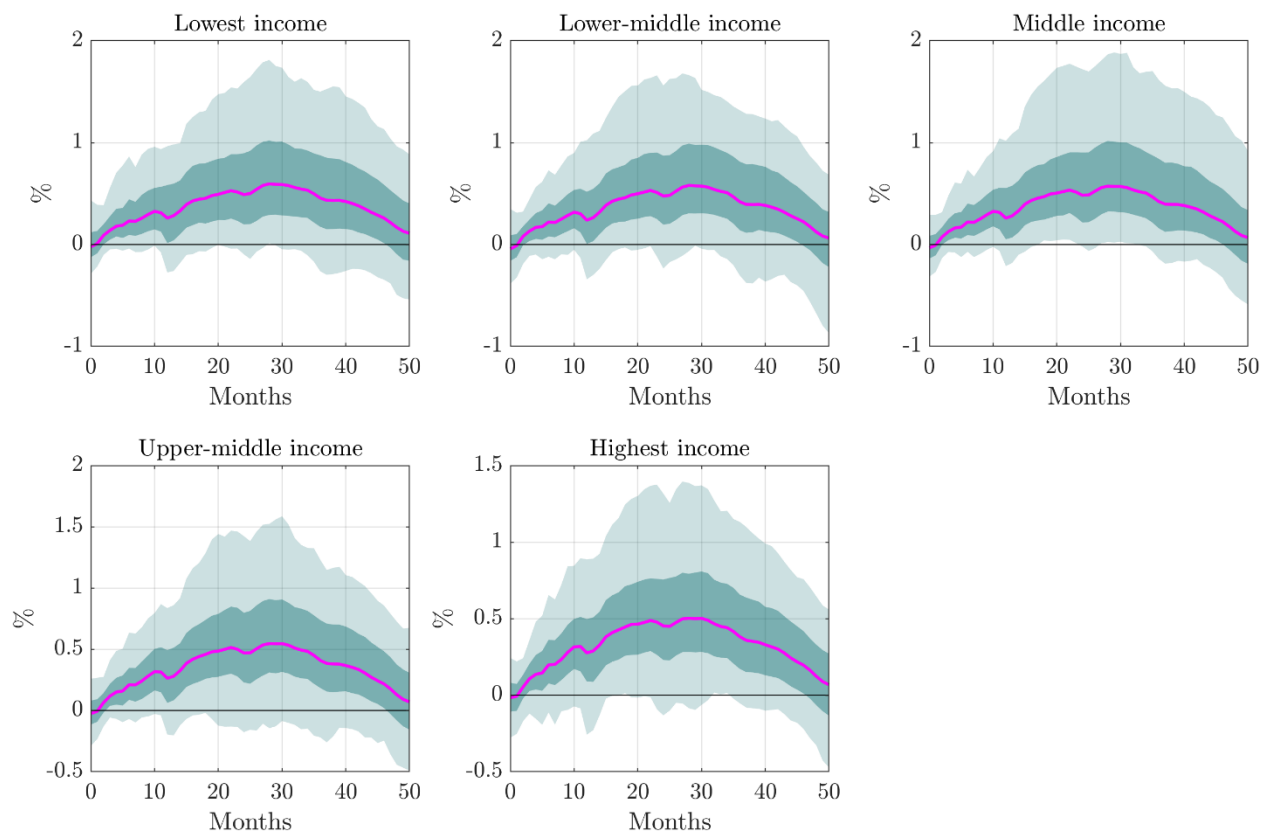
Notes: The shock is normalized to a 4 percent increase in the real biofuel price. Bands as in Figure 5. Impulse responses are plotted over a fifty-month horizon.

Appendix E: Distributional consequences

Table E.1. Summary statistics for food and electricity distributional CPIs by income quintile

Variable	Obs.	Mean	Std. Dev.	Min	Max
Lowest income	223	206.30	38.42	206.44	346.60
Lower-middle income	223	261.63	38.81	206.98	348.62
Middle income	223	263.23	39.48	207.77	351.17
Upper-middle income	223	264.36	39.97	208.27	353.06
Highest income	223	267.68	40.70	210.56	357.60

Figure E.1. Impulse responses of food and electricity distributional CPIs across income quintiles



Notes: The shock is normalized to a 4 percent increase in the real biofuel price. Bands as in Figure 5. Impulse responses are estimated by augmenting the baseline VAR with each biofuel price one at a time. Data are from the U.S. Bureau of Labor Statistics. We construct the food and electricity D-CPI based on Xavier Jaravel's replication code, published in his webpage (<https://www.xavierjaravel.com/dcpi>).